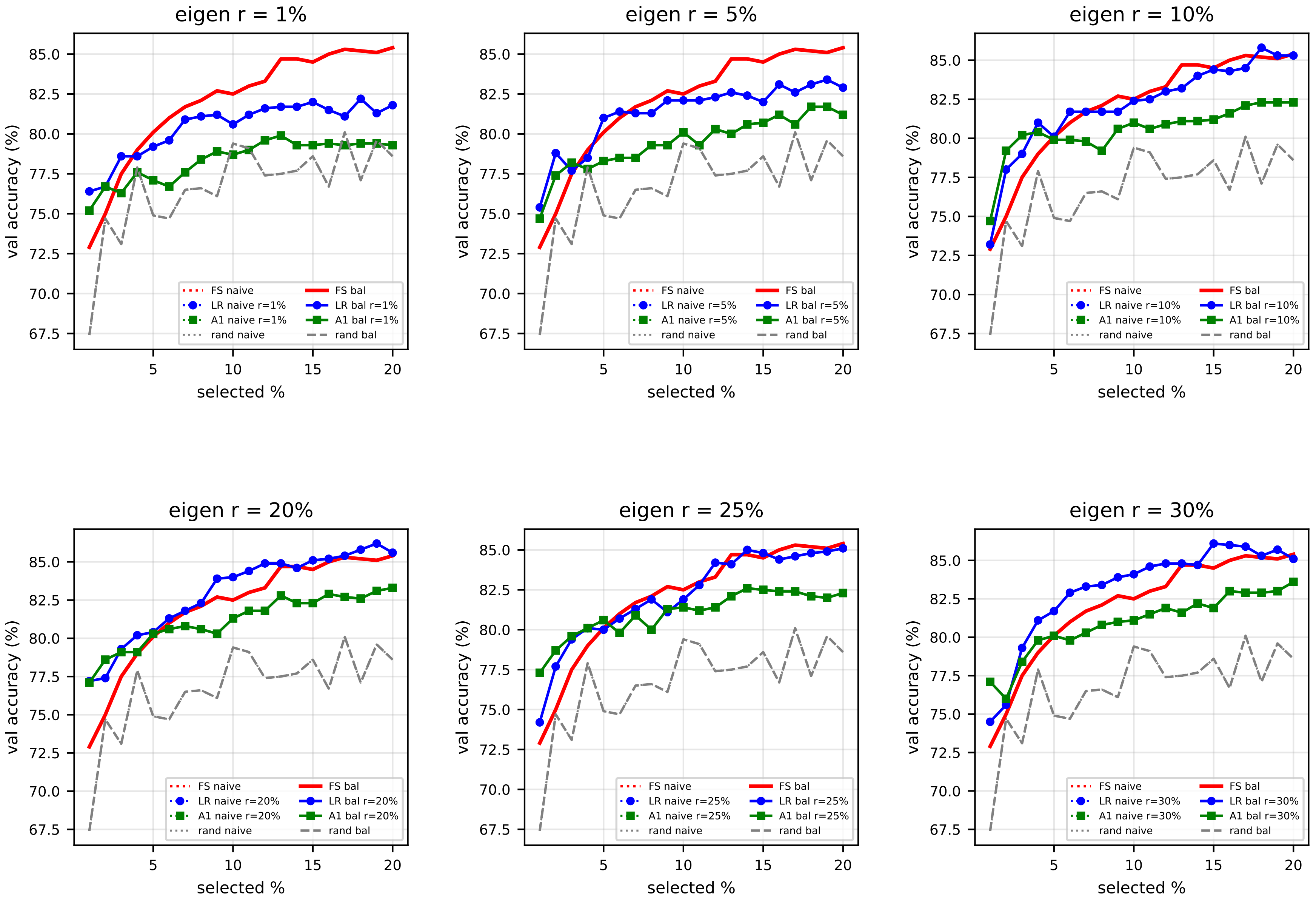
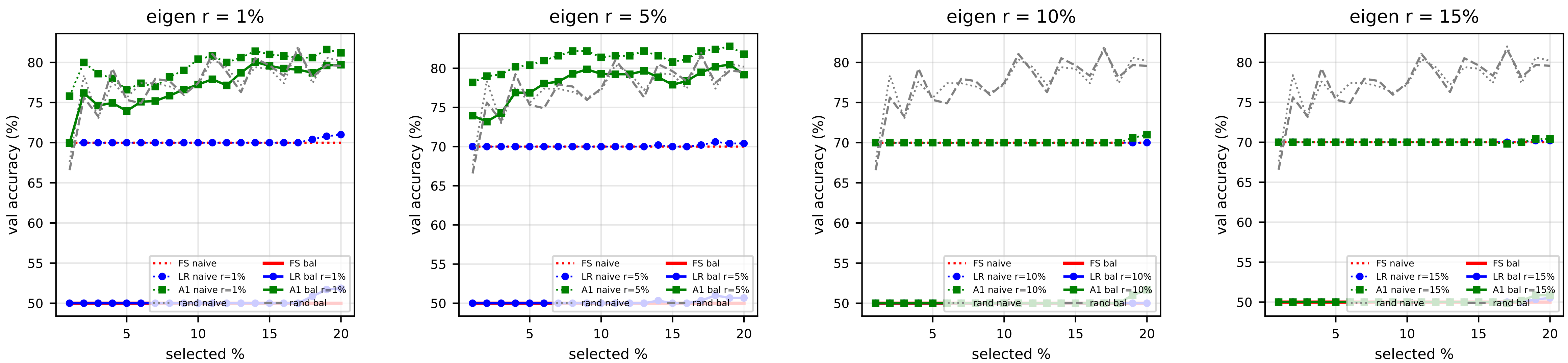
MR train 50/50 + val balanced 50/50 (n=4500, val=1000) | n_train=4500, val_tag=valbal1000, val_imb=balanced | metric = balanced accuracy

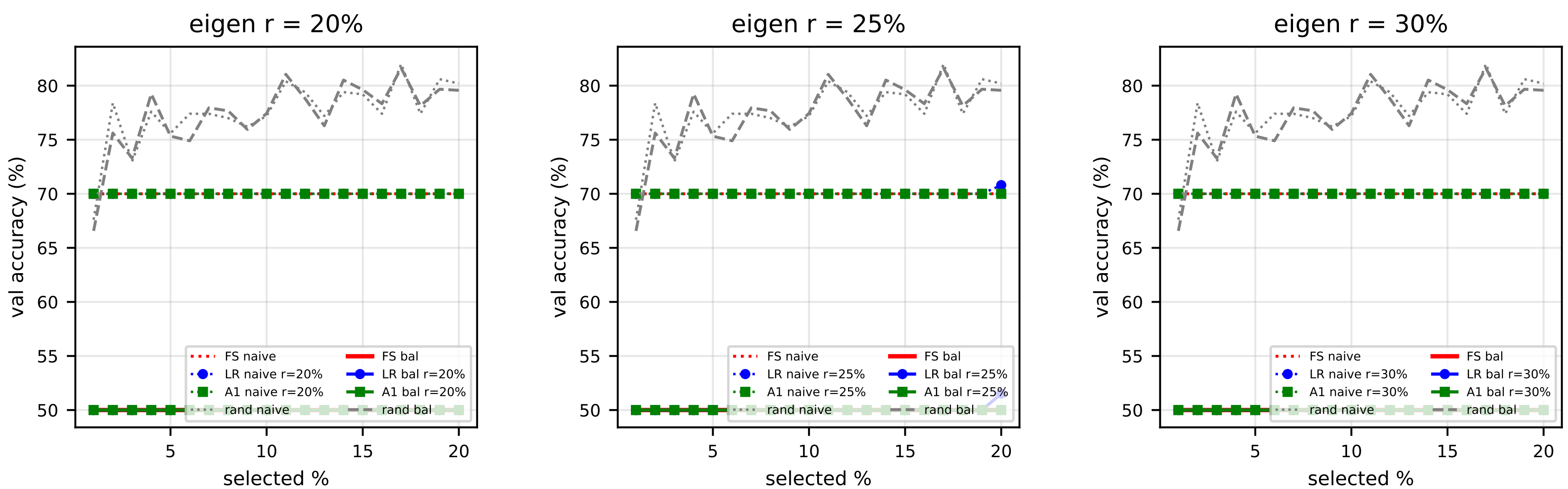



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (bal acc)	72.90	75.00	77.50	79.00	80.10	82.50	85.40
r=1%	X	-	X	!	X	!	X
r=5%	X	X	O	!	X	X	X
r=10%	O	O	O	X	X	X	X
r=15%	X	O	O	X	X	X	X
r=20%	X	O	X	X	X	X	X
r=25%	O	O	O	-	O	X	X
r=30%	O	O	X	X	X	X	X

MR train 50/50 + val 70/30 label1-maj (n=4500, val=500) | n_train=4500, val_tag=valimb500_pos70, val_imb=mild | metric = balanced accuracy

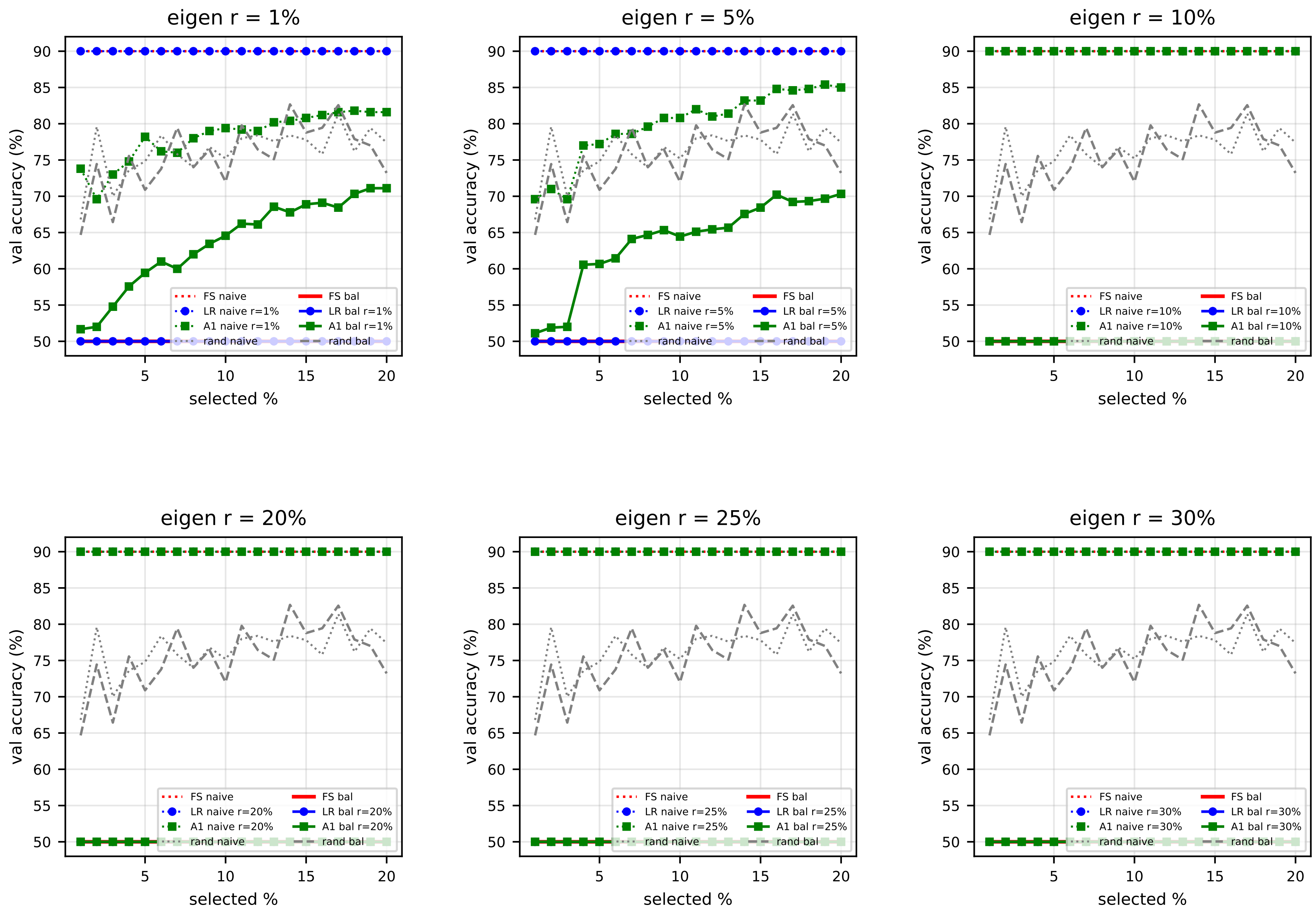





```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]
r=1%	O†	O†	O†	!†	!†	!†	O†
r=5%	O†	!†	O†	!†	O†	O†	!†
r=10%	!†	!†	!†	!†	!†	!†	!†
r=15%	!†	!†	!†	!†	!†	!†	!†
r=20%	!†	!†	!†	!†	!†	!†	!†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†

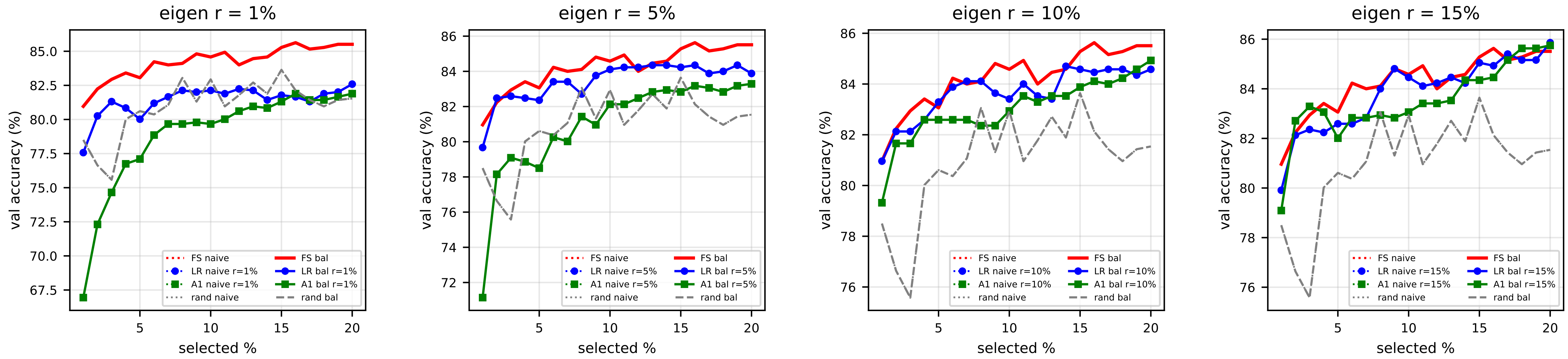
MR train 50/50 + val 90/10 label1-maj (n=4500, val=500) | n_train=4500, val_tag=valimb500_pos90, val_imb=extreme | metric = balanced accuracy

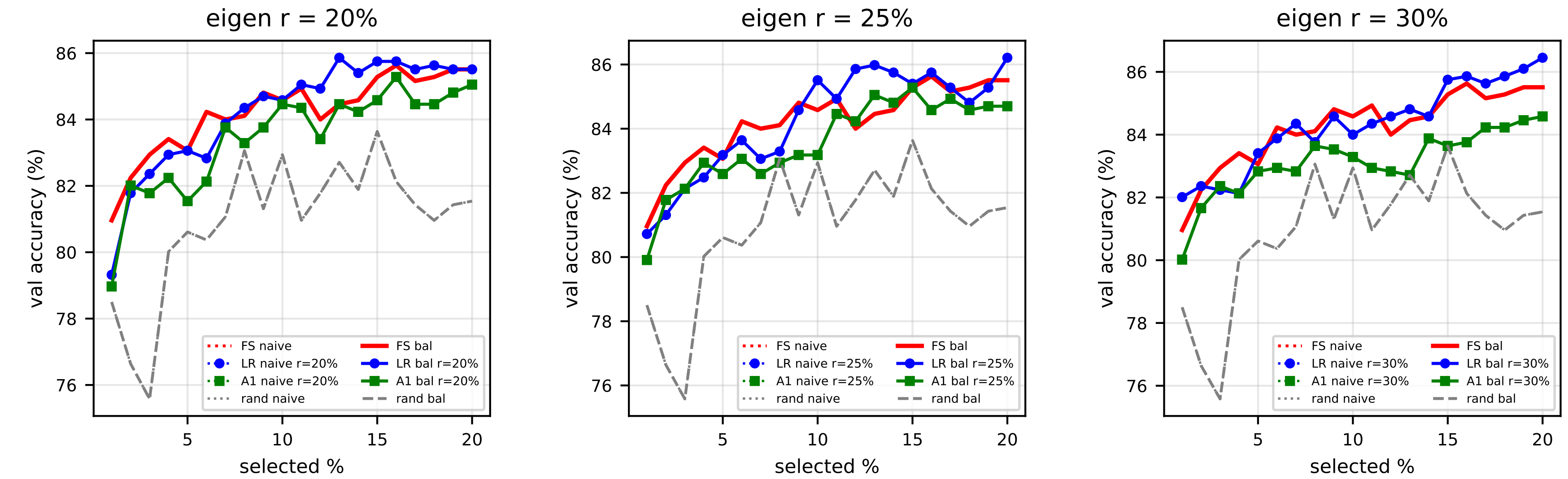



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]
r=1%	!†	!†	!†	!†	!†	!†	!†
r=5%	!†	!†	!†	!†	!†	!†	!†
r=10%	!†	!†	!†	!†	!†	!†	!†
r=15%	!†	!†	!†	!†	!†	!†	!†
r=20%	!†	!†	!†	!†	!†	!†	!†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†

SST-2 train 50/50 + val balanced 50/50 (n=5000, val=856) | n_train=5000, val_tag=valbal856, val_imb=balanced | metric = balanced accuracy





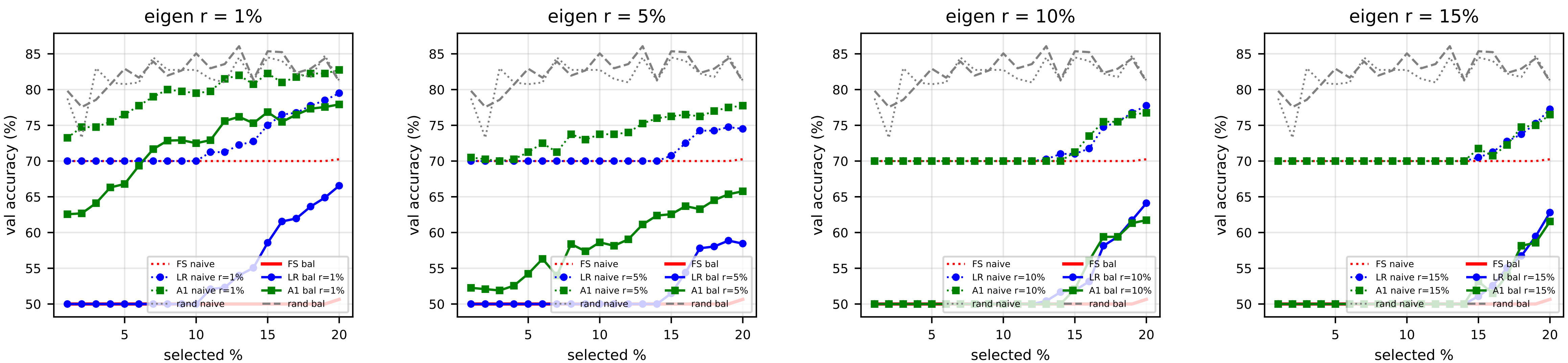
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

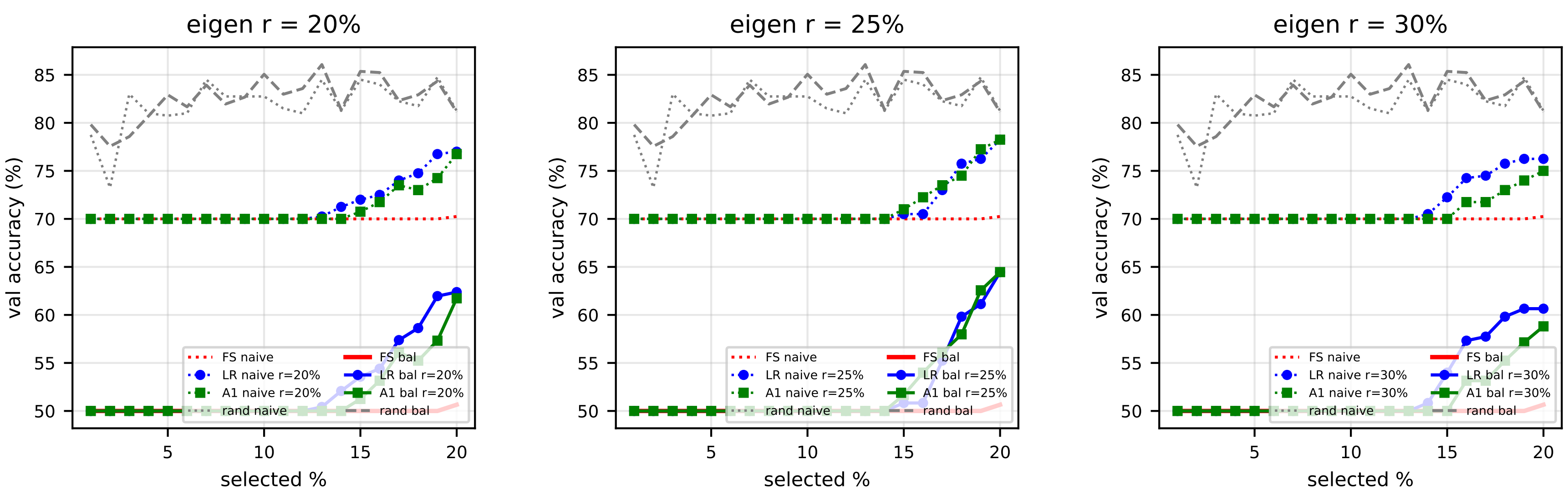
Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR (A1 also beats random)
  X : A1 < LR (A1 still beats random)
  - : A1 = LR (A1 still beats random)
  ! : A1 ≤ random (A1 itself harmful)
  † : LR ≤ random (LR collapsed –
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced]. Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	80.96	82.24	82.94	83.41	83.06	84.58	85.51
r=1%	!†	!	!	!	!†	!†	X
r=5%	!	X	X	!	!	!	X
r=10%	X	X	X	-	X	!	O
r=15%	X	O	O	O	X	X	X
r=20%	X	O	X	X	X	X	X
r=25%	X	O	-	O	X	X	X
r=30%	X	X	O	-	X	X	X

SST-2 train 50/50 + val 70/30 label1-maj (n=5000, val=400) | n_train=5000, val_tag=valimb400_pos70, val_imb=mild | metric = balanced accuracy

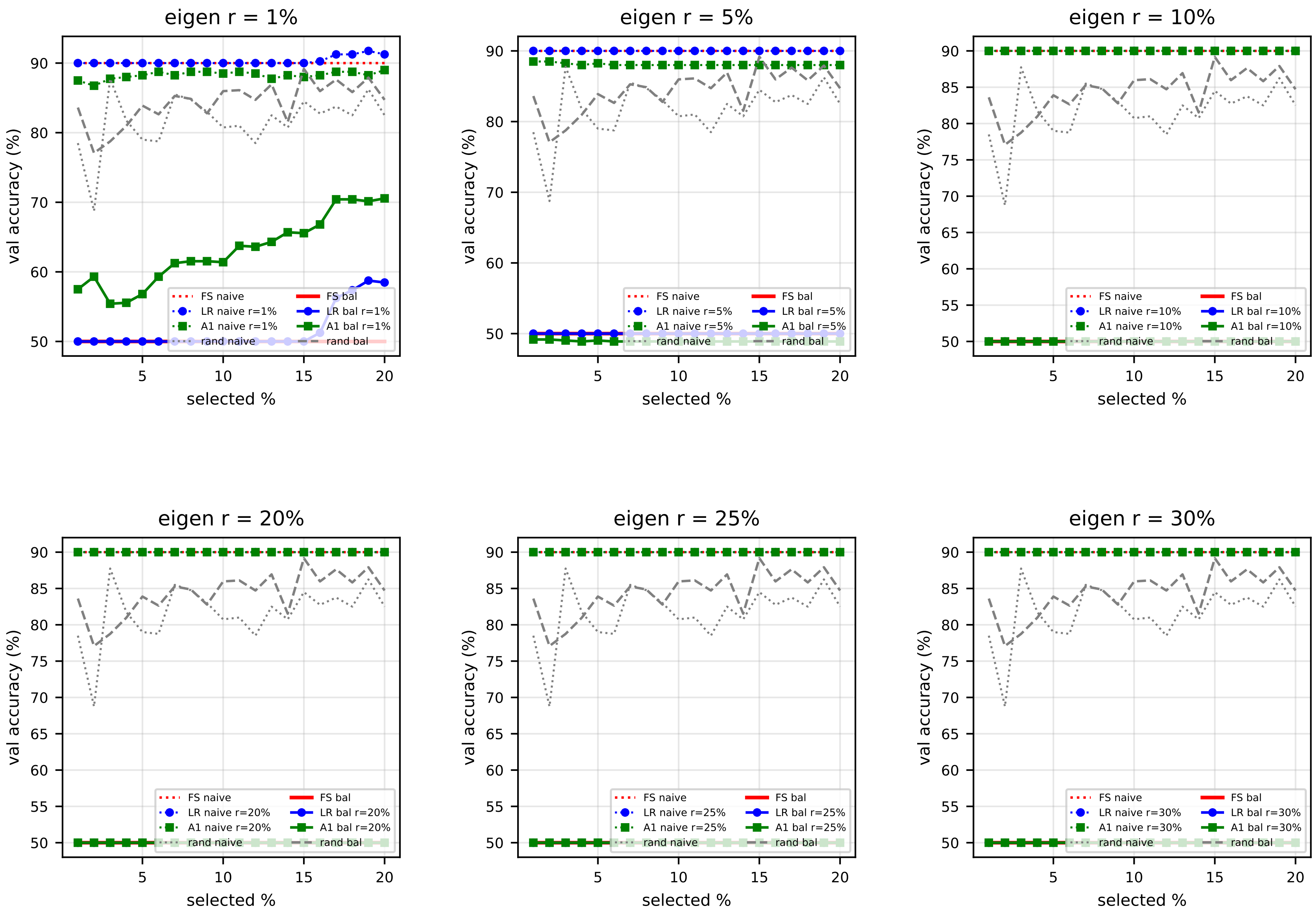





```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.65 [!]
r=1%	!†	!†	!†	!†	!†	!†	!†
r=5%	!†	!†	!†	!†	!†	!†	!†
r=10%	!†	!†	!†	!†	!†	!†	!†
r=15%	!†	!†	!†	!†	!†	!†	!†
r=20%	!†	!†	!†	!†	!†	!†	!†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†

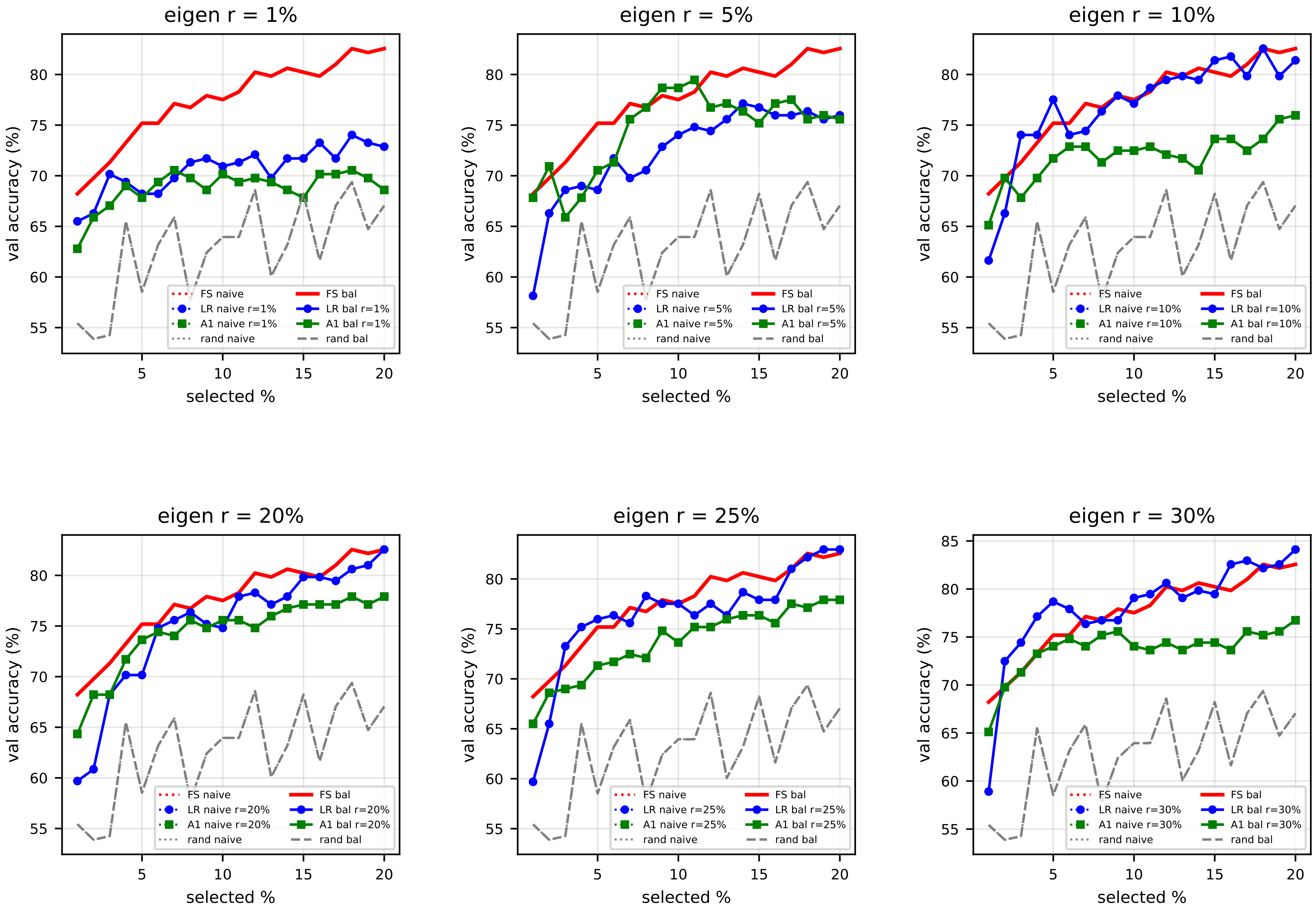
SST-2 train 50/50 + val 90/10 label1-maj (n=5000, val=400) | n_train=5000, val_tag=valimb400_pos90, val_imb=extreme | metric = balanced accuracy




```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]
r=1%	!†	!†	!†	!†	!†	!†	!†
r=5%	!†	!†	!†	!†	!†	!†	!†
r=10%	!†	!†	!†	!†	!†	!†	!†
r=15%	!†	!†	!†	!†	!†	!†	!†
r=20%	!†	!†	!†	!†	!†	!†	!†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†

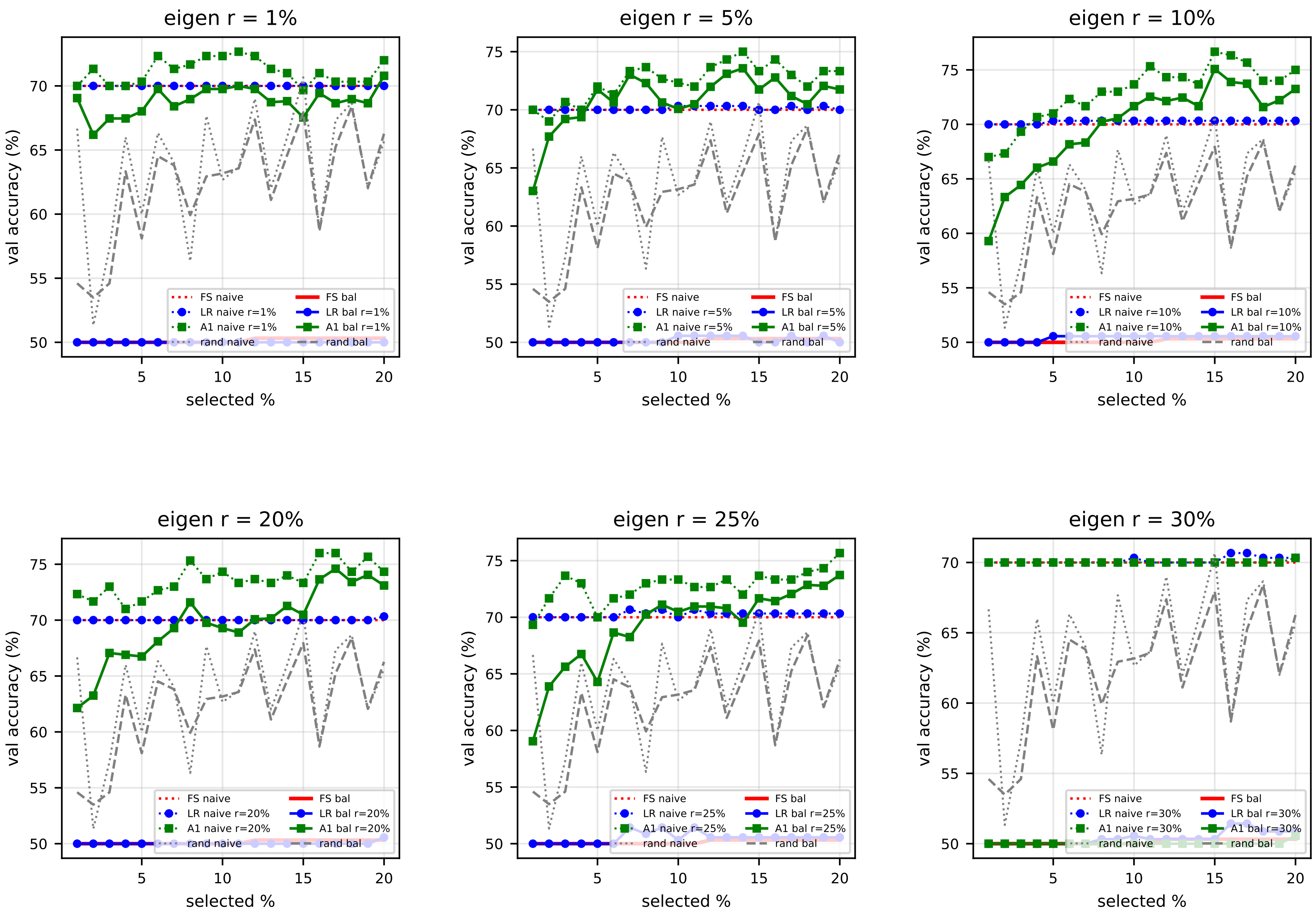
MRPC train 50/50 + val balanced 50/50 (n=2300, val=258) | n_train=2300, val_tag=valbal258, val_imb=balanced | metric = balanced accuracy




```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed –
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	68.22	69.77	71.32	73.26	75.19	77.52	82.56
r=1%	X	X	X	X	X	X	X
r=5%	O	O	X	X	O	O	X
r=10%	O	O	X	X	X	X	X
r=15%	X	O	X	X	-	X	X
r=20%	O	O	-	O	O	O	X
r=25%	O	O	X	X	X	X	X
r=30%	O	X	X	X	X	X	X

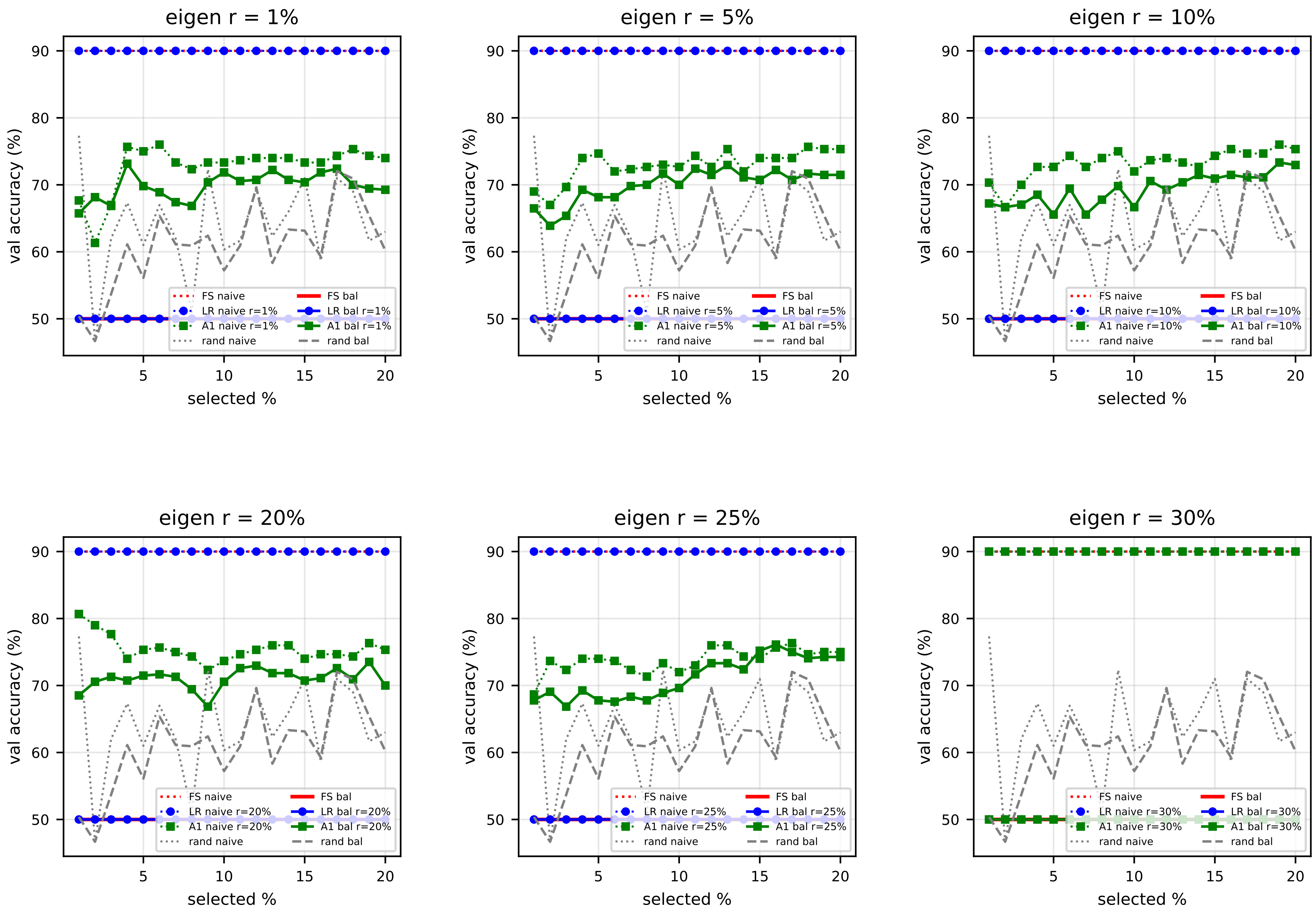
MRPC train 50/50 + val 70/30 label1-maj (n=2300, val=300) | n_train=2300, val_tag=valimb300_pos70, val_imb=mild | metric = balanced accuracy




```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.32 [!]
r=1%	O†	O†	O†	O†	O†	O†	O†
r=5%	O†	O†	O†	O†	O†	O†	O†
r=10%	O†	O†	O†	O†	O†	O†	O†
r=15%	O†	O†	O†	!†	O†	O†	O†
r=20%	O†	O†	O†	O†	O†	O†	O†
r=25%	O†	O†	O†	O†	O†	O†	O†
r=30%	!†	!†	!†	!†	!†	!†	!†

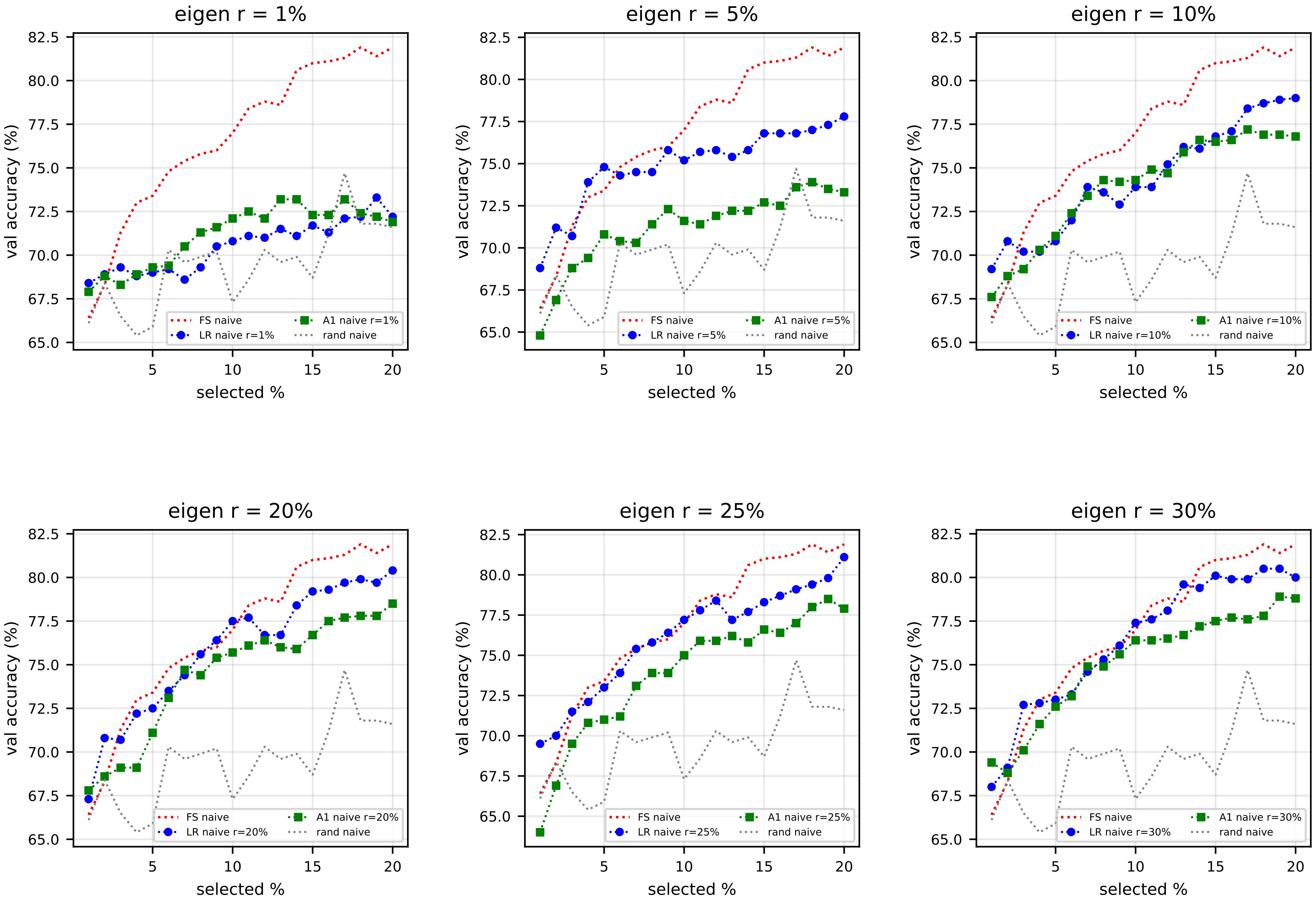
MRPC train 50/50 + val 90/10 label1-maj (n=2300, val=300) | n_train=2300, val_tag=valimb300_pos90, val_imb=extreme | metric = balanced accuracy




```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	50.00 [!]	50.00	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]
r=1%	O†	O	O†	O†	O†	O†	O†
r=5%	O†	O	O†	O†	O†	O†	O†
r=10%	O†	O	O†	O†	O†	O†	O†
r=15%	O†	O	O†	O†	O†	O†	O†
r=20%	O†	O	O†	O†	O†	O†	O†
r=25%	O†	O	O†	O†	O†	O†	O†
r=30%	!†	-	!†	!†	!†	!†	!†

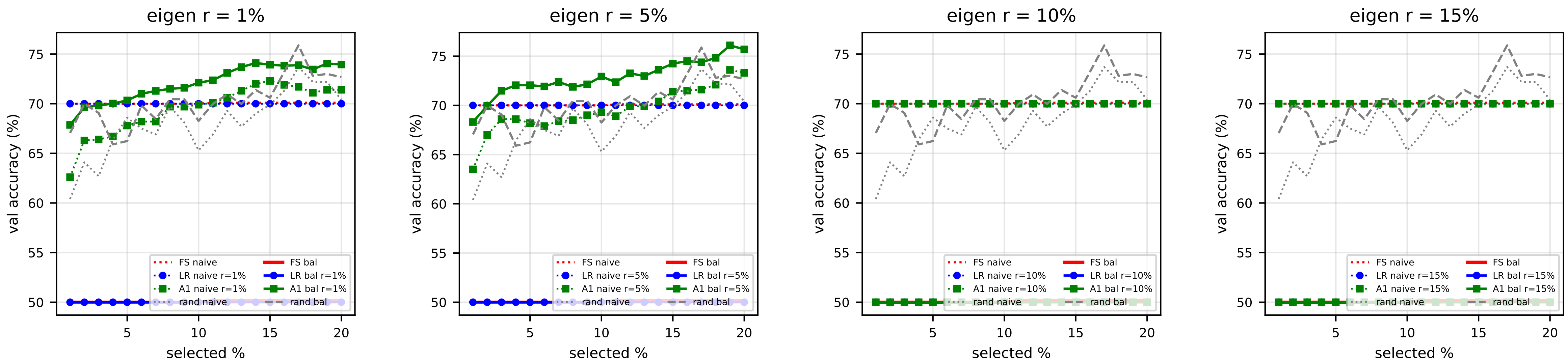
QQP train 50/50 + val balanced 50/50 (n=5000, val=1000) | n_train=5000, val_tag=valbal1000, val_imb=balanced | metric = balanced accuracy

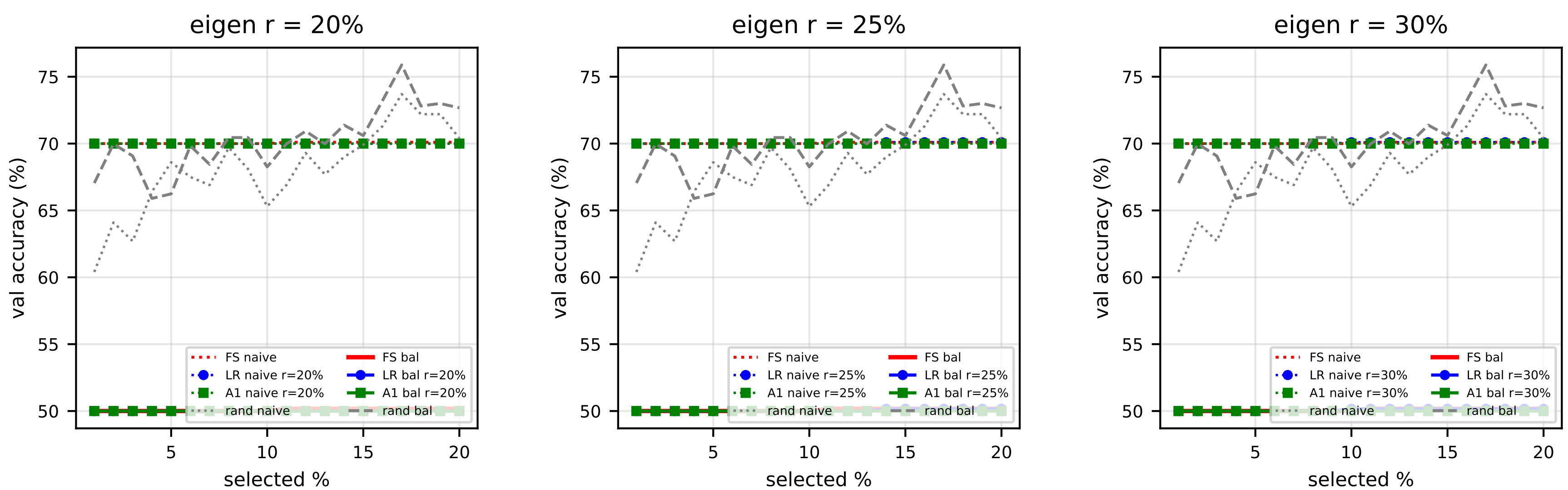



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed –
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

(balanced INV baseline missing — table unavailable)

QQP train 50/50 + val 70/30 label0-maj (n=5000, val=1000) | n_train=5000, val_tag=valimb1000_pos30, val_imb=mild | metric = balanced accuracy

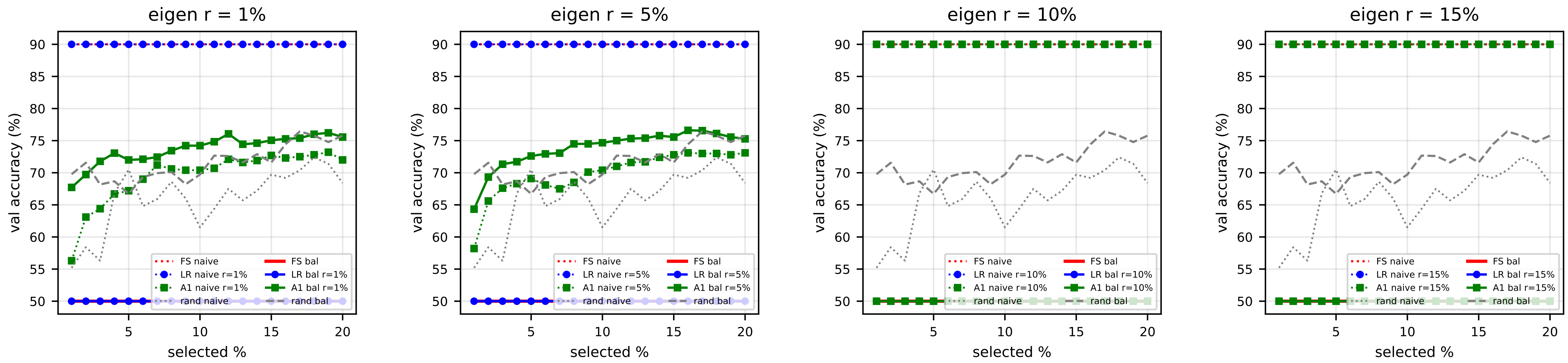


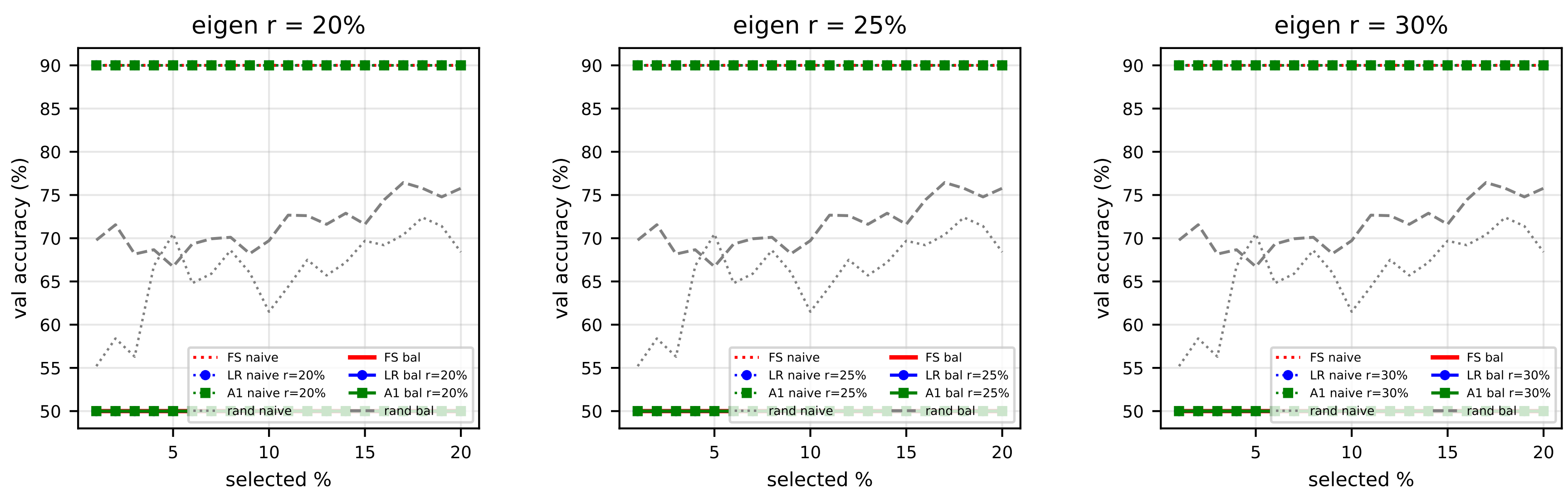



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.17 [!]
r=1%	O†	!†	O†	O†	O†	O†	O†
r=5%	O†	!†	O†	O†	O†	O†	O†
r=10%	!†	!†	!†	!†	!†	!†	!†
r=15%	!†	!†	!†	!†	!†	!†	!†
r=20%	!†	!†	!†	!†	!†	!†	!†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†

QQP train 50/50 + val 90/10 label0-maj (n=5000, val=1000) | n_train=5000, val_tag=valimb1000_pos10, val_imb=extreme | metric = balanced accuracy

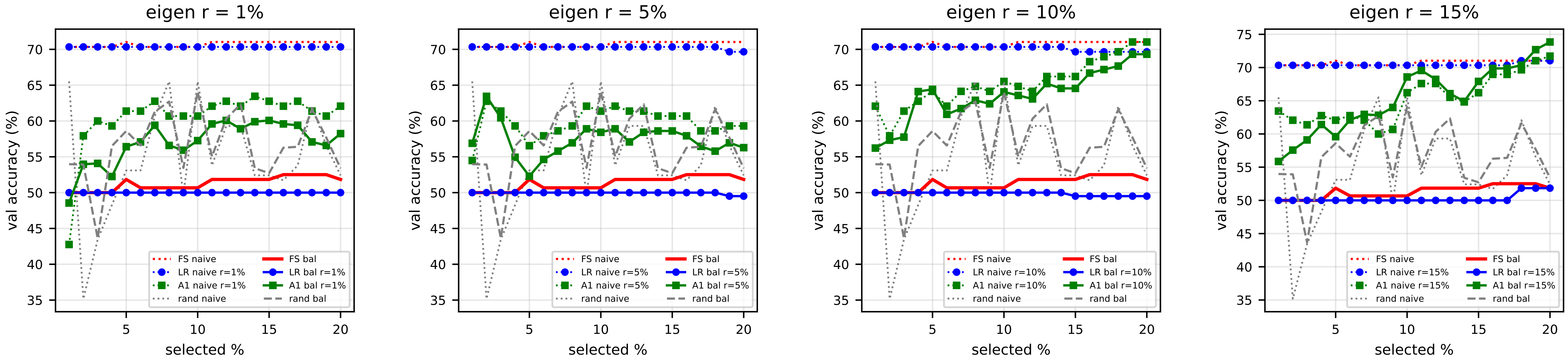


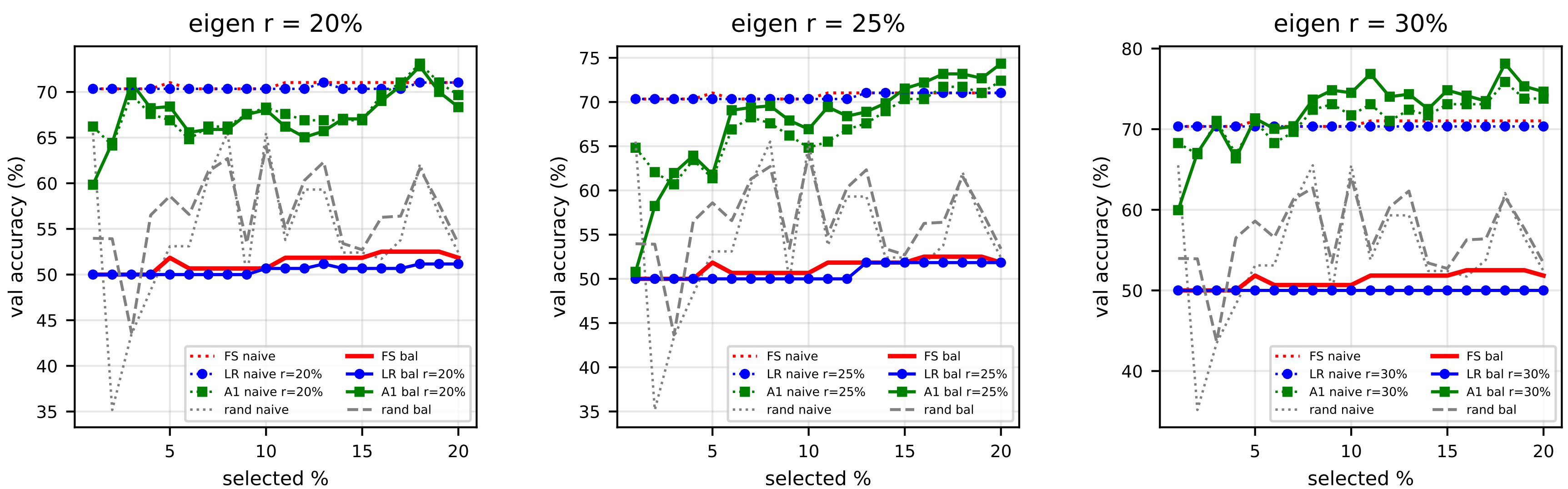



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed –
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]
r=1%	!†	!†	O†	O†	O†	O†	!†
r=5%	!†	!†	O†	O†	O†	O†	!†
r=10%	!†	!†	!†	!†	!†	!†	!†
r=15%	!†	!†	!†	!†	!†	!†	!†
r=20%	!†	!†	!†	!†	!†	!†	!†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†

RTE train 50/50 + val 70/30 label1-maj (n=1300, val=145) | n_train=1300, val_tag=valimb145_pos70, val_imb=mild | metric = balanced accuracy

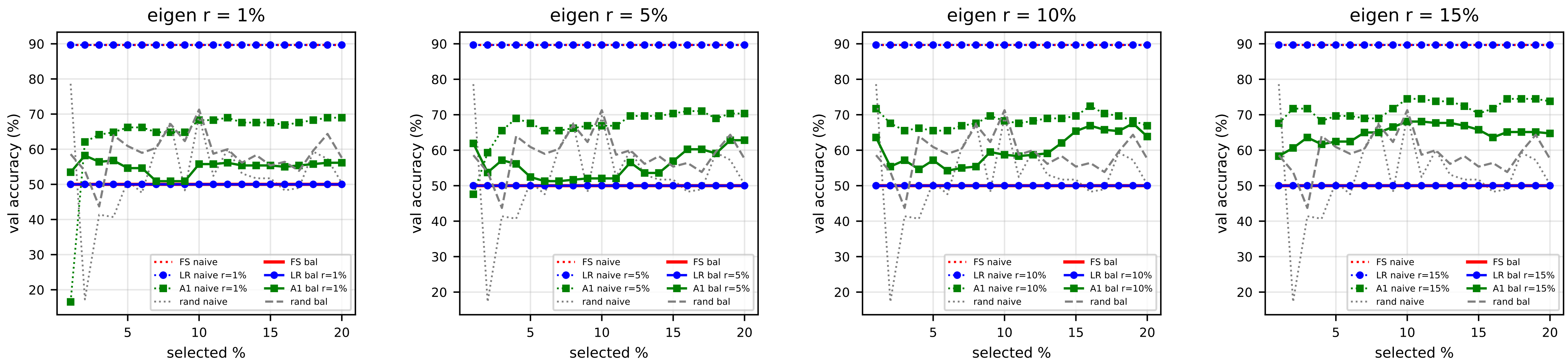


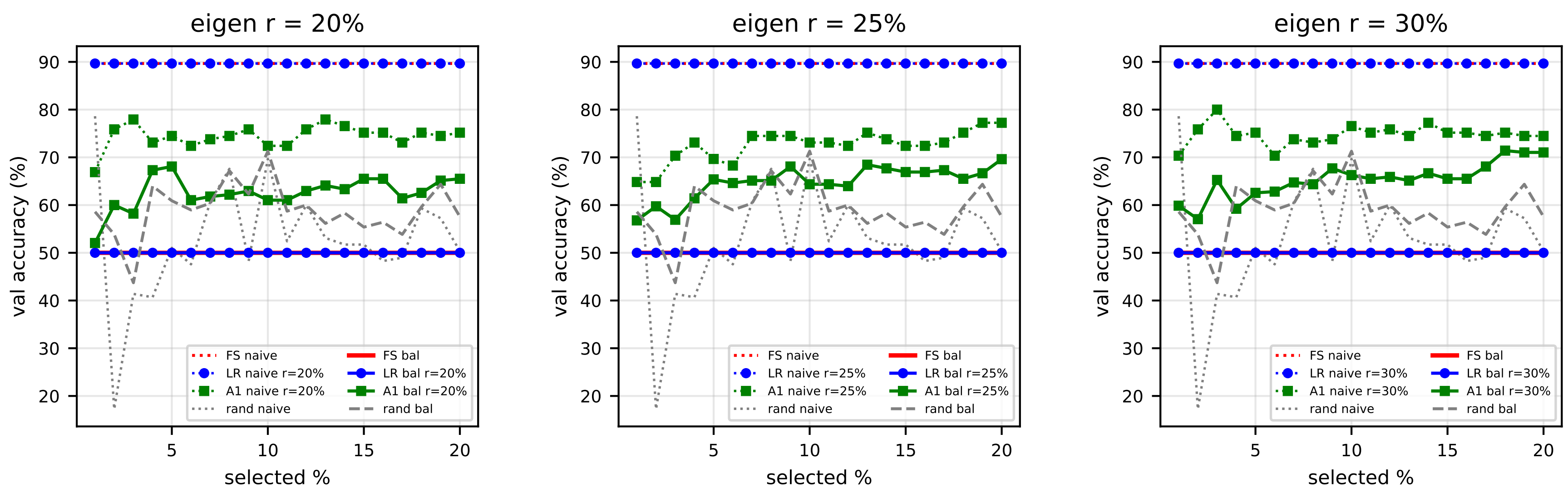



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed –
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (bal acc)	50.00 [!]	50.00 [!]	50.00	50.00 [!]	51.84 [!]	50.67 [!]	51.84 [!]
r=1%	!†	O†	O	!†	!†	!†	O†
r=5%	O†	O†	O	!†	!†	!†	O†
r=10%	O†	O†	O	O†	O†	!†	O†
r=15%	O†	O†	O	O†	O†	O†	O†
r=20%	O†	O†	O	O†	O†	O†	O†
r=25%	!†	O†	O	O†	O†	O†	O†
r=30%	O†	O†	O	O†	O†	O†	O†

RTE train 50/50 + val 90/10 label1-maj (n=1300, val=145) | n_train=1300, val_tag=valimb145_pos90, val_imb=extreme | metric = balanced accuracy

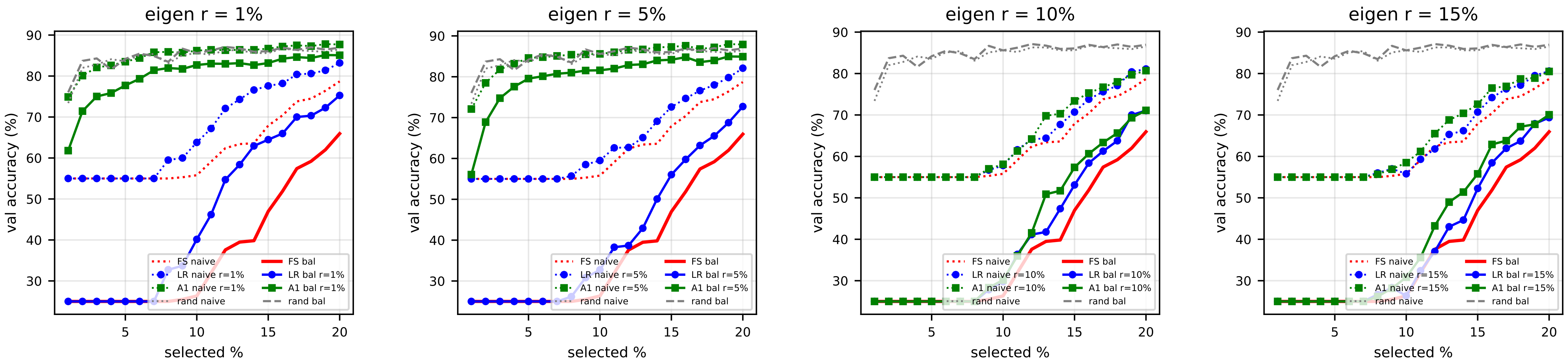


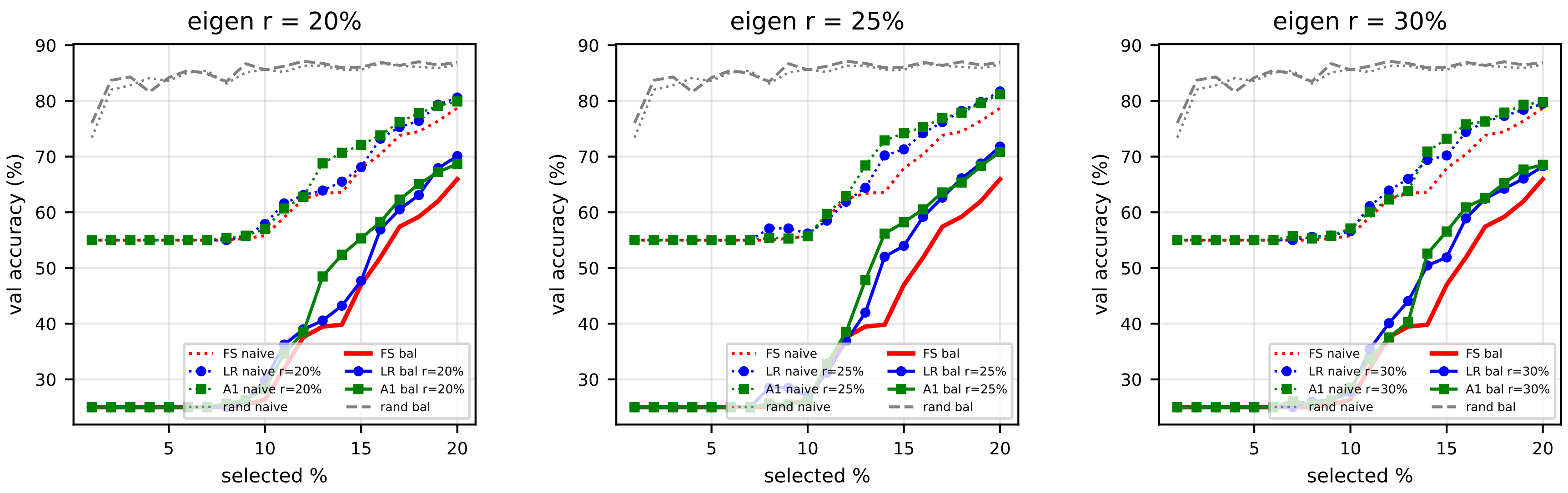



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (bal acc)	50.00 [!]	50.00 [!]	50.00	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]
r=1%	!†	O†	O	!†	!†	!†	!†
r=5%	O†	!†	O	!†	!†	!†	O†
r=10%	O†	O†	O	!†	!†	!†	O†
r=15%	!†	O†	O	!†	O†	!†	O†
r=20%	!†	O†	O	O†	O†	!†	O†
r=25%	!†	O†	O	!†	O†	!†	O†
r=30%	O†	O†	O	!†	O†	!†	O†

AG News train cls25 + val cls55_15_15_15 label0-maj (n=5000, val=1000) | n_train=5000, val_tag=valimb1000_cls55_15_15_15, val_imb=mild | metric = balanced accuracy

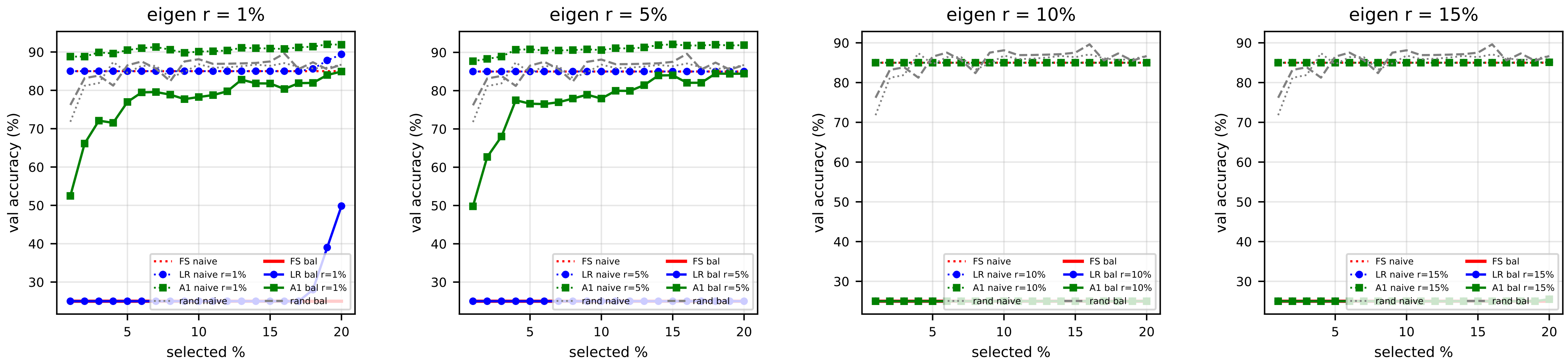


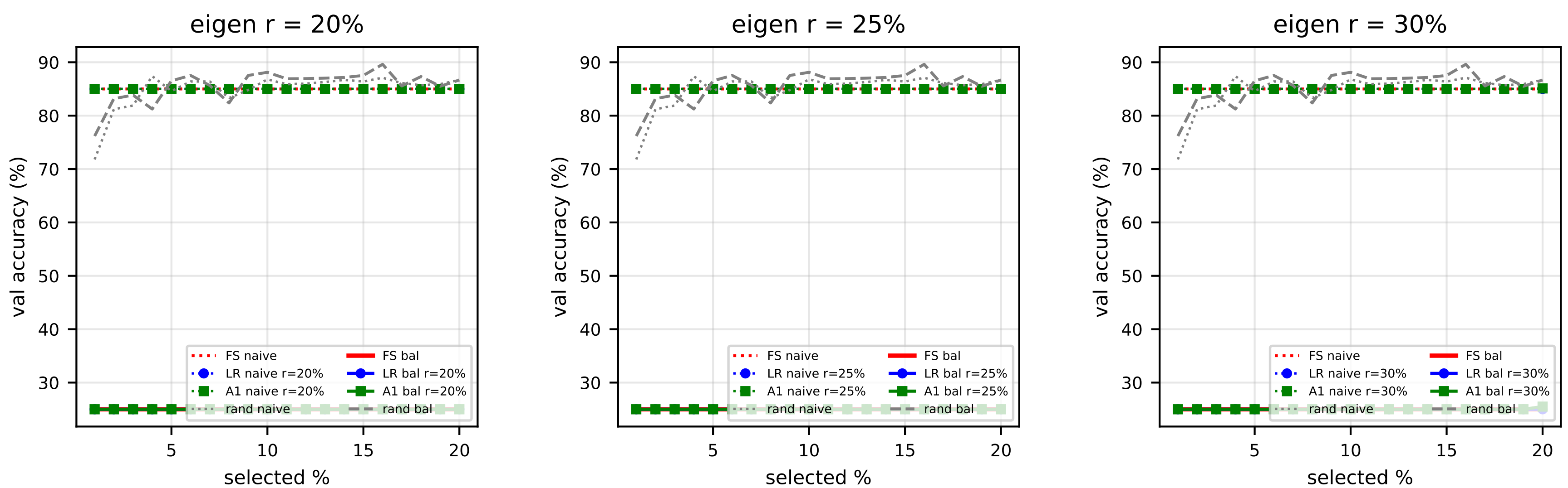



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed –
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	25.00 [!]	25.00 [!]	25.00 [!]	25.00 [!]	25.00 [!]	26.33 [!]	65.95 [!]
r=1%	!†	!†	!†	!†	!†	!†	!†
r=5%	!†	!†	!†	!†	!†	!†	!†
r=10%	!†	!†	!†	!†	!†	!†	!†
r=15%	!†	!†	!†	!†	!†	!†	!†
r=20%	!†	!†	!†	!†	!†	!†	!†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†

AG News train cls25 + val cls85_05_05_05 label0-maj (n=5000, val=1000) | n_train=5000, val_tag=valimb1000_cls85_05_05_05, val_imb=extreme | metric = balanced accuracy

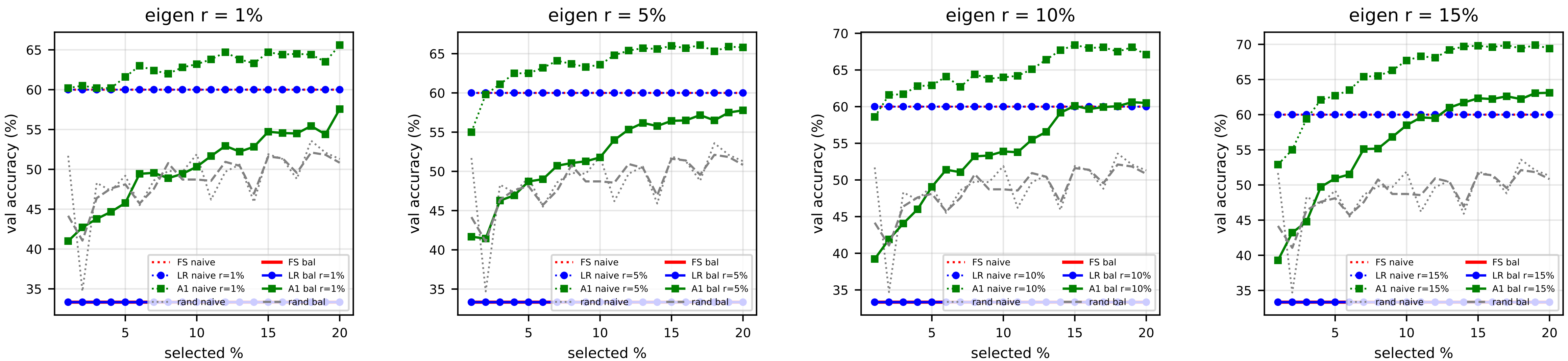


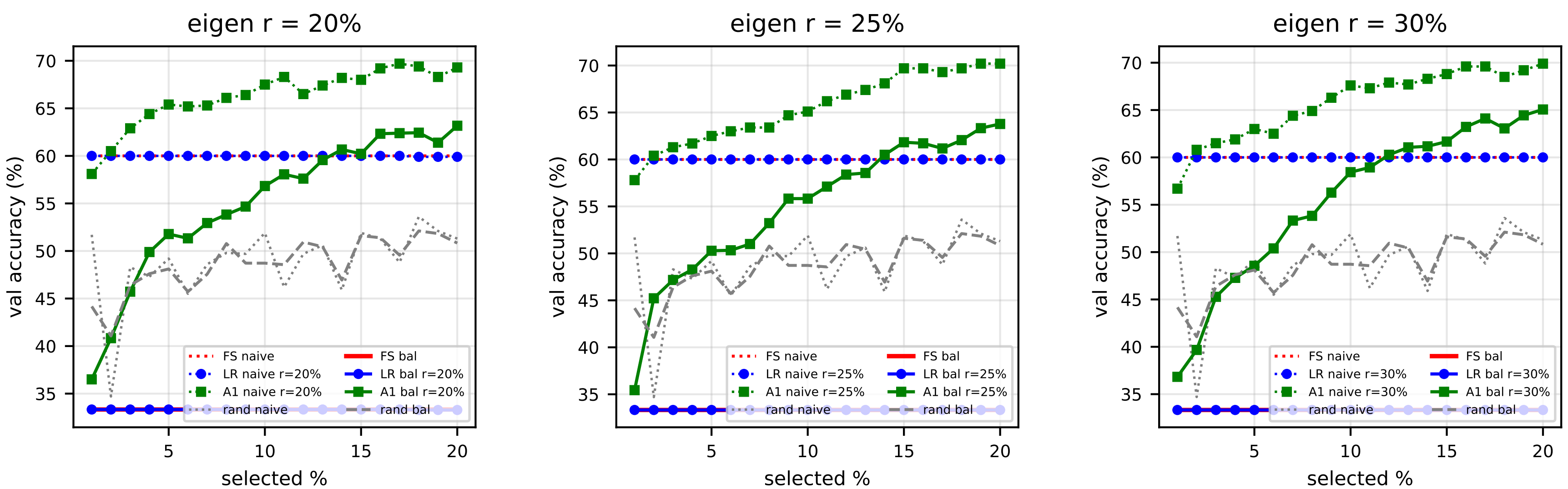



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	25.00 [!]	25.00 [!]	25.00 [!]	25.00 [!]	25.00 [!]	25.00 [!]	25.00 [!]
r=1%	!†	!†	!†	!†	!†	!†	!†
r=5%	!†	!†	!†	!†	!†	!†	!†
r=10%	!†	!†	!†	!†	!†	!†	!†
r=15%	!†	!†	!†	!†	!†	!†	!†
r=20%	!†	!†	!†	!†	!†	!†	!†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†

MNLI train cls33 + val cls60_20_20 label0-maj (n=5000, val=1000) | n_train=5000, val_tag=valimb1000_cls60_20_20, val_imb=mild | metric = balanced accuracy

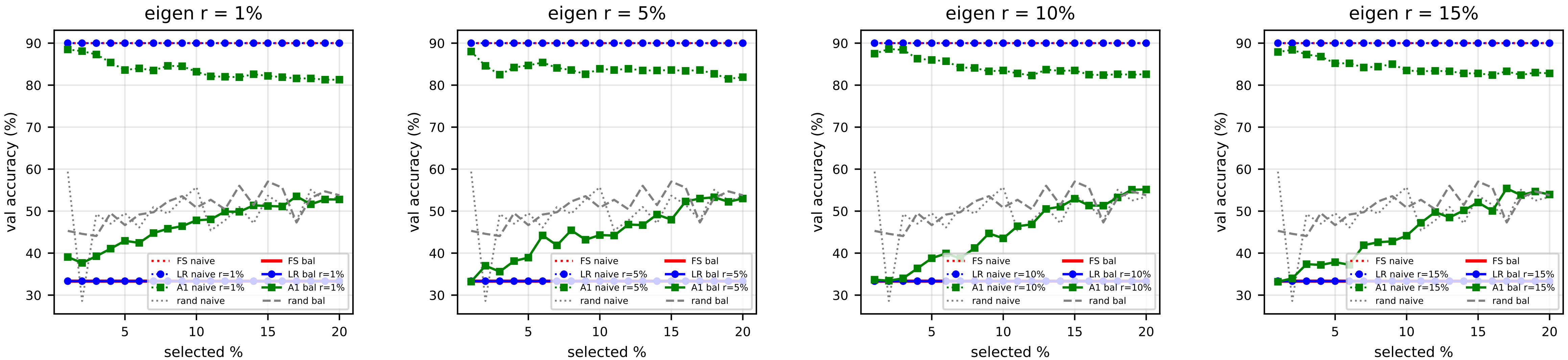


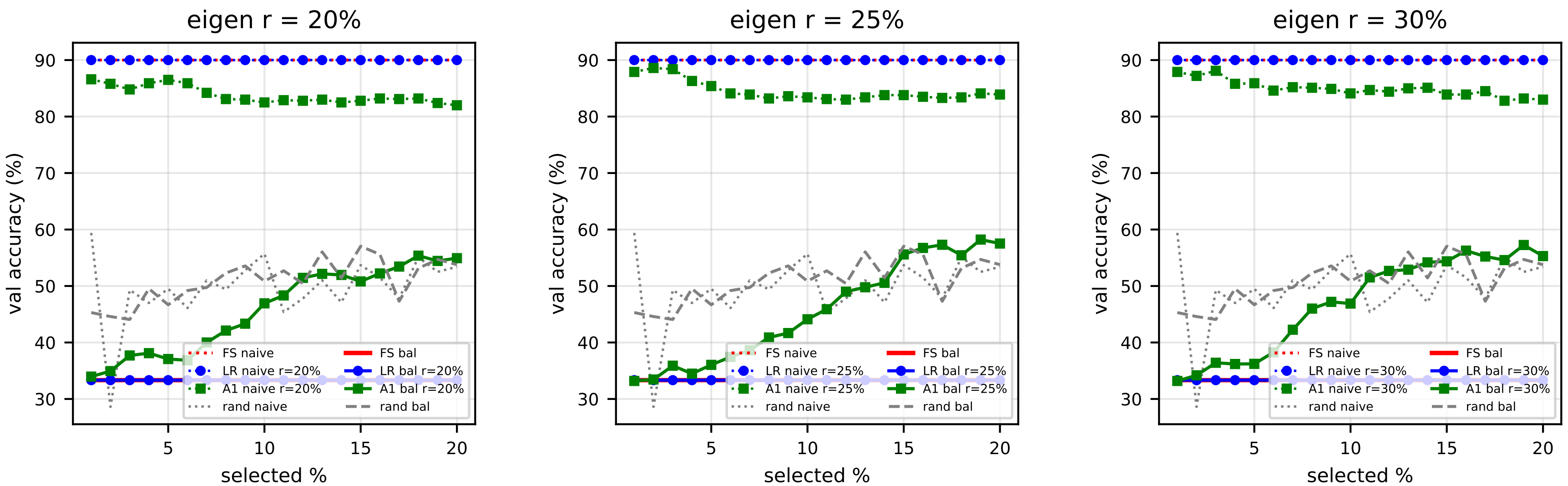



```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	33.33 [!]	33.33 [!]	33.33 [!]	33.33 [!]	33.33 [!]	33.33 [!]	33.33 [!]
r=1%	!†	O†	!†	!†	!†	O†	O†
r=5%	!†	O†	!†	!†	O†	O†	O†
r=10%	!†	O†	!†	!†	O†	O†	O†
r=15%	!†	O†	!†	O†	O†	O†	O†
r=20%	!†	!†	!†	O†	O†	O†	O†
r=25%	!†	O†	O†	O†	O†	O†	O†
r=30%	!†	!†	!†	!†	O†	O†	O†

MNLI train cls33 + val cls90_05_05 label0-maj (n=5000, val=1000) | n_train=5000, val_tag=valimb1000_cls90_05_05, val_imb=extreme | metric = balanced accuracy






```
PRIMARY metric: BALANCED accuracy (solid).
Naive accuracy shown for reference (dotted).

Markers in lower table (balanced acc, A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing _balanced data

Random baseline (figure curve) = FS-INV
random_results_inv[_balanced].  Missing
_balanced data is skipped point-by-point
(no curve drawn for that metric).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (bal acc)	33.33 [!]	33.33 [!]	33.33 [!]	33.33 [!]	33.33 [!]	33.33 [!]	33.33 [!]
r=1%	!†	!†	!†	!†	!†	!†	!†
r=5%	!†	!†	!†	!†	!†	!†	!†
r=10%	!†	!†	!†	!†	!†	!†	O†
r=15%	!†	!†	!†	!†	!†	!†	O†
r=20%	!†	!†	!†	!†	!†	!†	O†
r=25%	!†	!†	!†	!†	!†	!†	O†
r=30%	!†	!†	!†	!†	!†	!†	O†

Summary (BALANCED acc): A1 vs LRFShap vs random (per-cell breakdown)

Breakdown by setting (rows = dataset/ratio/val_tag; cols aggregate over rank × sel%)

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harn	A1 ≤ rnd (harm	A1 win %	LR-safe %	A1-safe %
mr/pos50/valbal1000	49	13	31	2	0	3	26.5	100.0	93.9
mr/pos50/valimb500_pos70	49	8	0	0	49	41	16.3	0.0	16.3
mr/pos50/valimb500_pos90	49	0	0	0	49	49	0.0	0.0	0.0
sst2/pos50/valbal856	49	8	27	3	3	11	16.3	93.9	77.6
sst2/pos50/valimb400_pos70	49	0	0	0	49	49	0.0	0.0	0.0
sst2/pos50/valimb400_pos90	49	0	0	0	49	49	0.0	0.0	0.0
mrpc/pos50/valbal258	49	15	32	2	0	0	30.6	100.0	100.0
mrpc/pos50/valimb300_pos70	49	41	0	0	49	8	83.7	0.0	83.7
mrpc/pos50/valimb300_pos90	49	42	0	1	42	6	85.7	14.3	87.8
qqp/pos50/valbal1000	0	0	0	0	0	0	-	-	-
qqp/pos50/valimb1000_pos30	49	12	0	0	49	37	24.5	0.0	24.5
qqp/pos50/valimb1000_pos10	49	8	0	0	49	41	16.3	0.0	16.3
rte/pos50/valimb145_pos70	49	40	0	0	42	9	81.6	14.3	81.6
rte/pos50/valimb145_pos90	49	27	0	0	42	22	55.1	14.3	55.1
news/cls25_25_25_25/valimb1000_cls55	49	0	0	0	49	49	0.0	0.0	0.0
news/cls25_25_25_25/valimb1000_cls85	49	0	0	0	49	49	0.0	0.0	0.0
mnli/cls33_33_33/valimb1000_cls60_2	49	29	0	0	49	20	59.2	0.0	59.2
mnli/cls33_33_33/valimb1000_cls90_0	49	5	0	0	49	44	10.2	0.0	10.2
ALL (18 settings)	**833**	**248**	**90**	**8**	**668**	**487**	**29.8**	**19.8**	**41.5**

Breakdown by rank (rows = eigen rank r%; aggregated across all settings × sel%)

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harn	A1 ≤ rnd (harn	A1 win %	LR-safe %	A1-safe %
rank r=1%	119	36	12	1	98	70	30.3	17.6	41.2
rank r=5%	119	44	11	0	95	64	37.0	20.2	46.2
rank r=10%	119	35	13	1	95	70	29.4	20.2	41.2
rank r=15%	119	36	14	1	95	68	30.3	20.2	42.9
rank r=20%	119	38	13	1	95	67	31.9	20.2	43.7
rank r=25%	119	39	11	2	95	67	32.8	20.2	43.7
rank r=30%	119	20	16	2	95	81	16.8	20.2	31.9
ALL ranks (18 settings)	**833**	**248**	**90**	**8**	**668**	**487**	**29.8**	**19.8**	**41.5**

```
Notes:
  - PRIMARY metric: balanced accuracy (top_results_inv_balanced).
  - n_cells = #valid (rank, sel%) pairs per setting (drops cells with
    missing _balanced data — affects qqp/valbal1000 entirely).
  - 'A1 win %' = #cells where A1 strictly beats LR (and beats random — '!' excluded).
  - 'LR-safe %' = #cells where LR > random; 'A1-safe %' = #cells where A1 > random.
  - Per-cell baseline = FS-INV balanced random (random_results_inv_balanced).
```

Per-rank breakdown grouped by val imbalance level (balanced / mild / extreme) — BALANCED acc

BALANCED imbalance — settings: mr/pos50/valbal1000, sst2/pos50/valbal856, mrpc/pos50/valbal258, qqp/pos50/valbal1000

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harm	A1 ≤ rnd (harm	A1 win %	LR-safe %	A1-safe %
r=1%	21	0	12	1	3	8	0.0	85.7	61.9
r=5%	21	5	11	0	0	5	23.8	100.0	76.2
r=10%	21	6	13	1	0	1	28.6	100.0	95.2
r=15%	21	6	14	1	0	0	28.6	100.0	100.0
r=20%	21	7	13	1	0	0	33.3	100.0	100.0
r=25%	21	8	11	2	0	0	38.1	100.0	100.0
r=30%	21	4	16	1	0	0	19.0	100.0	100.0
ALL ranks (4 settings)	**147**	**36**	**90**	**7**	**3**	**14**	**24.5**	**98.0**	**90.5**

MILD imbalance — settings: mr/pos50/valimb500_pos70, sst2/pos50/valimb400_pos70, mrpc/pos50/valimb300_pos70, qqp/pos50/valimb1000_pos30, rte/pos50/valimb145_pos70, ag_news/cls25_25_25_25/valimb1000_cls55_15_15_15, mnli/cls33_33_33/valimb1000_cls60_20_20

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harm	A1 ≤ rnd (harm	A1 win %	LR-safe %	A1-safe %
r=1%	49	23	0	0	48	26	46.9	2.0	46.9
r=5%	49	25	0	0	48	24	51.0	2.0	51.0
r=10%	49	17	0	0	48	32	34.7	2.0	34.7
r=15%	49	18	0	0	48	31	36.7	2.0	36.7
r=20%	49	18	0	0	48	31	36.7	2.0	36.7
r=25%	49	19	0	0	48	30	38.8	2.0	38.8
r=30%	49	10	0	0	48	39	20.4	2.0	20.4
ALL ranks (7 settings)	**343**	**130**	**0**	**0**	**336**	**213**	**37.9**	**2.0**	**37.9**

EXTREME imbalance — settings: mr/pos50/valimb500_pos90, sst2/pos50/valimb400_pos90, mrpc/pos50/valimb300_pos90, qqp/pos50/valimb1000_pos10, rte/pos50/valimb145_pos90, ag_news/cls25_25_25_25/valimb1000_cls85_05_05_05, mnli/cls33_33_33/valimb1000_cls90_05_05

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harm	A1 ≤ rnd (harm	A1 win %	LR-safe %	A1-safe %
r=1%	49	13	0	0	47	36	26.5	4.1	26.5
r=5%	49	14	0	0	47	35	28.6	4.1	28.6
r=10%	49	12	0	0	47	37	24.5	4.1	24.5
r=15%	49	12	0	0	47	37	24.5	4.1	24.5
r=20%	49	13	0	0	47	36	26.5	4.1	26.5
r=25%	49	12	0	0	47	37	24.5	4.1	24.5
r=30%	49	6	0	1	47	42	12.2	4.1	14.3
ALL ranks (7 settings)	**343**	**82**	**0**	**1**	**329**	**260**	**23.9**	**4.1**	**24.2**

```
Group definitions in this trainbal+valvary fork:
  - BALANCED: val_tag = valbal*    (val composition matches train 50/50)
  - MILD:     val_tag = valimb*_pos70  (val label1 70%)
  - EXTREME:  val_tag = valimb*_pos90  (val label1 90%)

Each row aggregates 7 sel% × all settings in that group at the given rank.
Random baseline = FS-INV balanced random.  qqp/valbal1000 contributes
0 cells because its sidecar predates the balanced-acc rerun.
```

Indicator: A1 > LR (balanced) (cell = #settings satisfying / #valid settings)

group / rank	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%	ALL sel%
r=1%	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/21
r=5%	1/3	1/3	1/3	0/3	1/3	1/3	0/3	5/21
r=10%	2/3	2/3	1/3	0/3	0/3	0/3	1/3	6/21
r=15%	0/3	3/3	2/3	1/3	0/3	0/3	0/3	6/21
r=20%	1/3	3/3	0/3	1/3	1/3	1/3	0/3	7/21
r=25%	2/3	3/3	1/3	1/3	1/3	0/3	0/3	8/21
r=30%	2/3	1/3	1/3	0/3	0/3	0/3	0/3	4/21
BALANCED ALL (n=4)	8/21	13/21	6/21	3/21	3/21	2/21	1/21	36/147
r=1%	6/7	7/7	7/7	7/7	7/7	7/7	7/7	48/49
r=5%	7/7	7/7	7/7	7/7	7/7	7/7	7/7	49/49
r=10%	3/7	3/7	3/7	3/7	3/7	4/7	5/7	24/49
r=15%	3/7	3/7	3/7	3/7	3/7	4/7	5/7	24/49
r=20%	3/7	3/7	3/7	3/7	3/7	3/7	3/7	21/49
r=25%	3/7	3/7	3/7	3/7	3/7	3/7	3/7	21/49
r=30%	2/7	2/7	2/7	2/7	2/7	3/7	3/7	16/49
MILD ALL (n=7)	27/49	28/49	28/49	28/49	28/49	31/49	33/49	203/343
r=1%	7/7	7/7	7/7	7/7	7/7	7/7	7/7	49/49
r=5%	5/7	6/7	6/7	6/7	6/7	6/7	6/7	41/49
r=10%	3/7	3/7	3/7	3/7	3/7	3/7	3/7	21/49
r=15%	2/7	3/7	3/7	3/7	3/7	3/7	3/7	20/49
r=20%	3/7	3/7	3/7	3/7	3/7	3/7	3/7	21/49
r=25%	2/7	3/7	3/7	3/7	3/7	3/7	3/7	20/49
r=30%	1/7	2/7	2/7	2/7	2/7	2/7	3/7	14/49
EXTREME ALL (n=7)	23/49	27/49	27/49	27/49	27/49	27/49	28/49	186/343

```
Reading guide (BALANCED accuracy):
  - Each cell is '<numerator>/<denominator>' where the numerator counts how
    many settings in that group satisfy `A1 > LR  (balanced)` at that (rank, sel%).
  - Denominator drops settings whose _balanced sidecar key is missing
    at that cell, so qqp/valbal1000 (no _balanced) is dropped everywhere
    and the partial mrpc cells drop one column each.
  - 'ALL sel%' sums numerators / denominators across the 7 sel% columns.
  - The '<GROUP> ALL (n=N)' row sums over that group's N settings and
    the 7 ranks for each sel%.  Trainbal+valvary fork has 4 BALANCED
    (mr/sst2/mrpc/qqp; qqp drops out), 3 MILD (mr/sst2/mrpc), and 3
    EXTREME (mr/sst2/mrpc) settings.
```

Indicator: A1 > INV (full kernel, balanced) (cell = #settings satisfying / #valid settings)

group / rank	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%	ALL sel%
r=1%	1/3	1/3	0/3	0/3	0/3	0/3	0/3	2/21
r=5%	1/3	2/3	1/3	0/3	0/3	1/3	0/3	5/21
r=10%	1/3	1/3	1/3	1/3	0/3	0/3	0/3	4/21
r=15%	1/3	3/3	2/3	1/3	0/3	0/3	1/3	8/21
r=20%	1/3	1/3	1/3	1/3	1/3	0/3	0/3	5/21
r=25%	1/3	1/3	1/3	1/3	1/3	0/3	0/3	5/21
r=30%	1/3	1/3	1/3	1/3	0/3	0/3	0/3	4/21
BALANCED ALL (n=4)	7/21	10/21	7/21	5/21	2/21	1/21	1/21	33/147
r=1%	6/7	7/7	7/7	7/7	7/7	7/7	7/7	48/49
r=5%	7/7	7/7	7/7	7/7	7/7	7/7	7/7	49/49
r=10%	3/7	3/7	3/7	3/7	3/7	4/7	6/7	25/49
r=15%	3/7	3/7	3/7	3/7	3/7	4/7	6/7	25/49
r=20%	3/7	3/7	3/7	3/7	3/7	4/7	5/7	24/49
r=25%	3/7	3/7	3/7	3/7	3/7	3/7	5/7	23/49
r=30%	2/7	2/7	2/7	2/7	2/7	3/7	5/7	18/49
MILD ALL (n=7)	27/49	28/49	28/49	28/49	28/49	32/49	41/49	212/343
r=1%	7/7	7/7	7/7	7/7	7/7	7/7	7/7	49/49
r=5%	5/7	6/7	6/7	6/7	6/7	6/7	6/7	41/49
r=10%	3/7	3/7	3/7	3/7	3/7	3/7	3/7	21/49
r=15%	2/7	3/7	3/7	3/7	3/7	3/7	4/7	21/49
r=20%	3/7	3/7	3/7	3/7	3/7	3/7	3/7	21/49
r=25%	2/7	3/7	3/7	3/7	3/7	3/7	3/7	20/49
r=30%	1/7	2/7	2/7	2/7	2/7	2/7	3/7	14/49
EXTREME ALL (n=7)	23/49	27/49	27/49	27/49	27/49	27/49	29/49	187/343

```
Reading guide (BALANCED accuracy):
  - Each cell is '<numerator>/<denominator>' where the numerator counts how
    many settings in that group satisfy `A1 > INV (full kernel, balanced)` at that (rank, sel%).
  - Denominator drops settings whose _balanced sidecar key is missing
    at that cell, so qqp/valbal1000 (no _balanced) is dropped everywhere
    and the partial mrpc cells drop one column each.
  - 'ALL sel%' sums numerators / denominators across the 7 sel% columns.
  - The '<GROUP> ALL (n=N)' row sums over that group's N settings and
    the 7 ranks for each sel%.  Trainbal+valvary fork has 4 BALANCED
    (mr/sst2/mrpc/qqp; qqp drops out), 3 MILD (mr/sst2/mrpc), and 3
    EXTREME (mr/sst2/mrpc) settings.
```

Indicator: A1 > random (balanced) (cell = #settings satisfying / #valid settings)

group / rank	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%	ALL sel%
r=1%	2/3	2/3	2/3	1/3	2/3	1/3	3/3	13/21
r=5%	2/3	3/3	3/3	1/3	2/3	2/3	3/3	16/21
r=10%	3/3	3/3	3/3	3/3	3/3	2/3	3/3	20/21
r=15%	3/3	3/3	3/3	3/3	3/3	3/3	3/3	21/21
r=20%	3/3	3/3	3/3	3/3	3/3	3/3	3/3	21/21
r=25%	3/3	3/3	3/3	3/3	3/3	3/3	3/3	21/21
r=30%	3/3	3/3	3/3	3/3	3/3	3/3	3/3	21/21
BALANCED ALL (n=4)	19/21	20/21	20/21	17/21	19/21	17/21	21/21	133/147
r=1%	3/7	4/7	4/7	2/7	2/7	3/7	5/7	23/49
r=5%	4/7	3/7	4/7	2/7	4/7	4/7	4/7	25/49
r=10%	2/7	3/7	2/7	2/7	3/7	2/7	3/7	17/49
r=15%	2/7	3/7	2/7	2/7	3/7	3/7	3/7	18/49
r=20%	2/7	2/7	2/7	3/7	3/7	3/7	3/7	18/49
r=25%	1/7	3/7	3/7	3/7	3/7	3/7	3/7	19/49
r=30%	1/7	1/7	1/7	1/7	2/7	2/7	2/7	10/49
MILD ALL (n=7)	15/49	19/49	18/49	15/49	20/49	20/49	23/49	130/343
r=1%	1/7	2/7	3/7	2/7	2/7	2/7	1/7	13/49
r=5%	2/7	1/7	3/7	2/7	2/7	2/7	2/7	14/49
r=10%	2/7	2/7	2/7	1/7	1/7	1/7	3/7	12/49
r=15%	1/7	2/7	2/7	1/7	2/7	1/7	3/7	12/49
r=20%	1/7	2/7	2/7	2/7	2/7	1/7	3/7	13/49
r=25%	1/7	2/7	2/7	1/7	2/7	1/7	3/7	12/49
r=30%	1/7	2/7	1/7	0/7	1/7	0/7	2/7	7/49
EXTREME ALL (n=7)	9/49	13/49	15/49	9/49	12/49	8/49	17/49	83/343

```
Reading guide (BALANCED accuracy):
  - Each cell is '<numerator>/<denominator>' where the numerator counts how
    many settings in that group satisfy `A1 > random  (balanced)` at that (rank, sel%).
  - Denominator drops settings whose _balanced sidecar key is missing
    at that cell, so qqp/valbal1000 (no _balanced) is dropped everywhere
    and the partial mrpc cells drop one column each.
  - 'ALL sel%' sums numerators / denominators across the 7 sel% columns.
  - The '<GROUP> ALL (n=N)' row sums over that group's N settings and
    the 7 ranks for each sel%.  Trainbal+valvary fork has 4 BALANCED
    (mr/sst2/mrpc/qqp; qqp drops out), 3 MILD (mr/sst2/mrpc), and 3
    EXTREME (mr/sst2/mrpc) settings.
```

Indicator: A1 > max(LR, INV, random) (balanced) (cell = #settings satisfying / #valid settings)

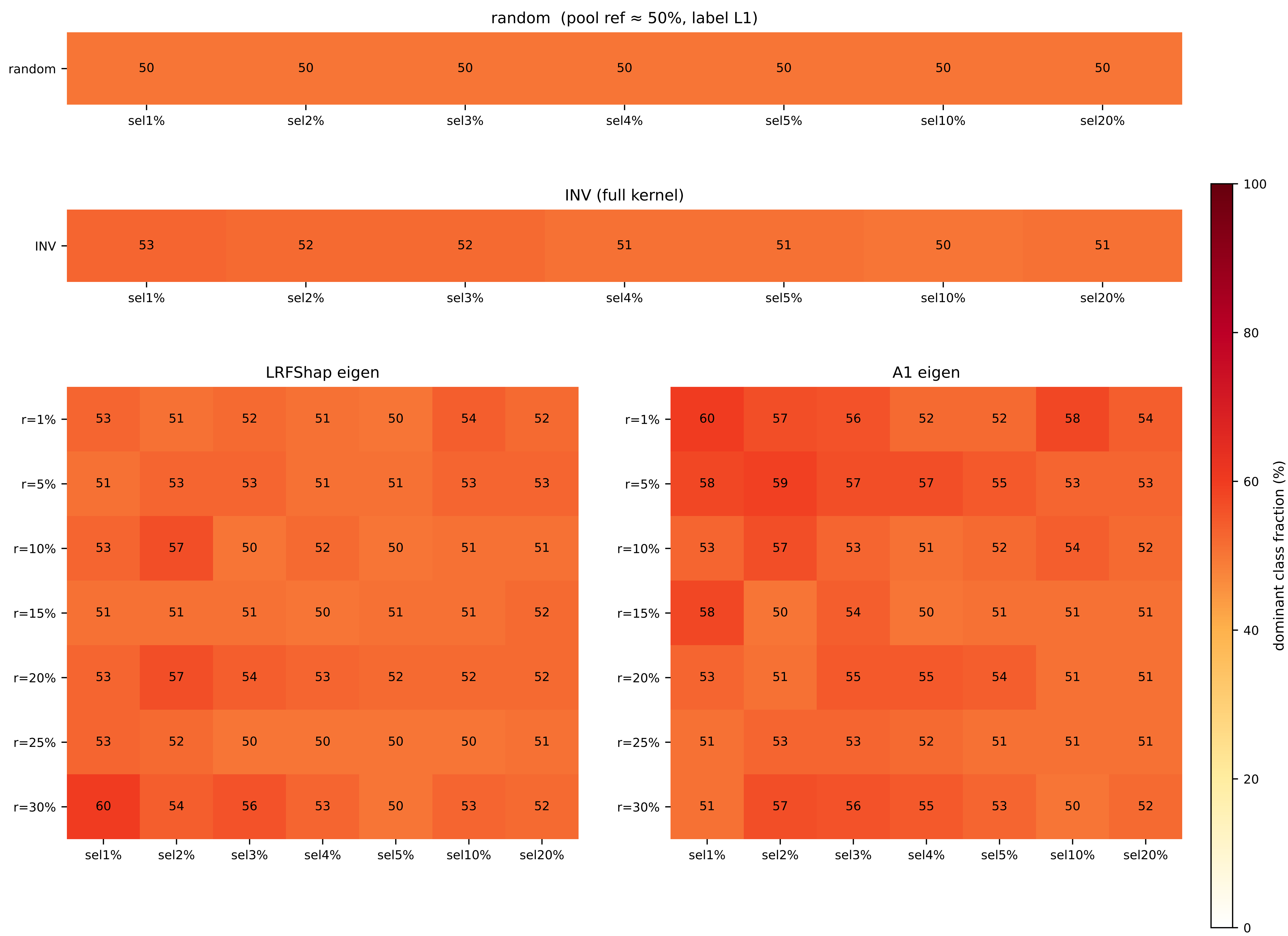
group / rank	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%	ALL sel%
r=1%	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/21
r=5%	0/3	1/3	1/3	0/3	0/3	1/3	0/3	3/21
r=10%	1/3	1/3	1/3	0/3	0/3	0/3	0/3	3/21
r=15%	0/3	3/3	2/3	0/3	0/3	0/3	0/3	5/21
r=20%	0/3	1/3	0/3	0/3	0/3	0/3	0/3	1/21
r=25%	1/3	1/3	1/3	0/3	1/3	0/3	0/3	4/21
r=30%	1/3	1/3	0/3	0/3	0/3	0/3	0/3	2/21
BALANCED ALL (n=4)	3/21	8/21	5/21	0/21	1/21	1/21	0/21	18/147
r=1%	3/7	4/7	4/7	2/7	2/7	3/7	5/7	23/49
r=5%	4/7	3/7	4/7	2/7	4/7	4/7	4/7	25/49
r=10%	2/7	3/7	2/7	2/7	3/7	2/7	3/7	17/49
r=15%	2/7	3/7	2/7	2/7	3/7	3/7	3/7	18/49
r=20%	2/7	2/7	2/7	3/7	3/7	3/7	3/7	18/49
r=25%	1/7	3/7	3/7	3/7	3/7	3/7	3/7	19/49
r=30%	1/7	1/7	1/7	1/7	2/7	2/7	2/7	10/49
MILD ALL (n=7)	15/49	19/49	18/49	15/49	20/49	20/49	23/49	130/343
r=1%	1/7	2/7	3/7	2/7	2/7	2/7	1/7	13/49
r=5%	2/7	1/7	3/7	2/7	2/7	2/7	2/7	14/49
r=10%	2/7	2/7	2/7	1/7	1/7	1/7	3/7	12/49
r=15%	1/7	2/7	2/7	1/7	2/7	1/7	3/7	12/49
r=20%	1/7	2/7	2/7	2/7	2/7	1/7	3/7	13/49
r=25%	1/7	2/7	2/7	1/7	2/7	1/7	3/7	12/49
r=30%	1/7	1/7	1/7	0/7	1/7	0/7	2/7	6/49
EXTREME ALL (n=7)	9/49	12/49	15/49	9/49	12/49	8/49	17/49	82/343

```
Reading guide (BALANCED accuracy):
  - Each cell is '<numerator>/<denominator>' where the numerator counts how
    many settings in that group satisfy `A1 > max(LR, INV, random)  (balanced)` at that (rank, sel%).
  - Denominator drops settings whose _balanced sidecar key is missing
    at that cell, so qqp/valbal1000 (no _balanced) is dropped everywhere
    and the partial mrpc cells drop one column each.
  - 'ALL sel%' sums numerators / denominators across the 7 sel% columns.
  - The '<GROUP> ALL (n=N)' row sums over that group's N settings and
    the 7 ranks for each sel%.  Trainbal+valvary fork has 4 BALANCED
    (mr/sst2/mrpc/qqp; qqp drops out), 3 MILD (mr/sst2/mrpc), and 3
    EXTREME (mr/sst2/mrpc) settings.
```

mr pos50 valbal1000 — top-Shapley class composition

MR train 50/50 + val balanced 50/50 (n=4500, val=1000)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L1=53 / L1=50	L1=52 / L1=50	L0=52 / L1=50	L1=51 / L1=50	L1=51 / L1=50	L1=50 / L1=50	L1=51 / L1=50
LR / A1							
r=1%	L0=53 / L0=60	L0=51 / L0=57	L0=52 / L0=56	L1=51 / L0=52	L1=50 / L0=52	L1=54 / L0=58	L1=52 / L0=54
r=5%	L1=51 / L0=58	L1=53 / L0=59	L1=53 / L0=57	L1=51 / L0=57	L1=51 / L0=55	L1=53 / L0=53	L1=53 / L0=53
r=10%	L0=53 / L0=53	L0=57 / L0=57	L0=50 / L0=53	L0=52 / L1=51	L1=50 / L1=52	L0=51 / L1=54	L1=51 / L1=52
r=15%	L0=51 / L1=58	L0=51 / L0=50	L1=51 / L1=54	L0=50 / L0=50	L0=51 / L1=51	L1=51 / L1=51	L1=52 / L1=51
r=20%	L0=53 / L1=53	L1=57 / L0=51	L1=54 / L1=55	L1=53 / L1=55	L1=52 / L1=54	L1=52 / L1=51	L1=52 / L1=51
r=25%	L0=53 / L0=51	L0=52 / L1=53	L1=50 / L1=53	L1=50 / L1=52	L0=50 / L1=51	L1=50 / L1=51	L1=51 / L1=51
r=30%	L0=60 / L0=51	L0=54 / L1=57	L0=56 / L1=56	L0=53 / L1=55	L1=50 / L1=53	L1=53 / L0=50	L1=52 / L1=52

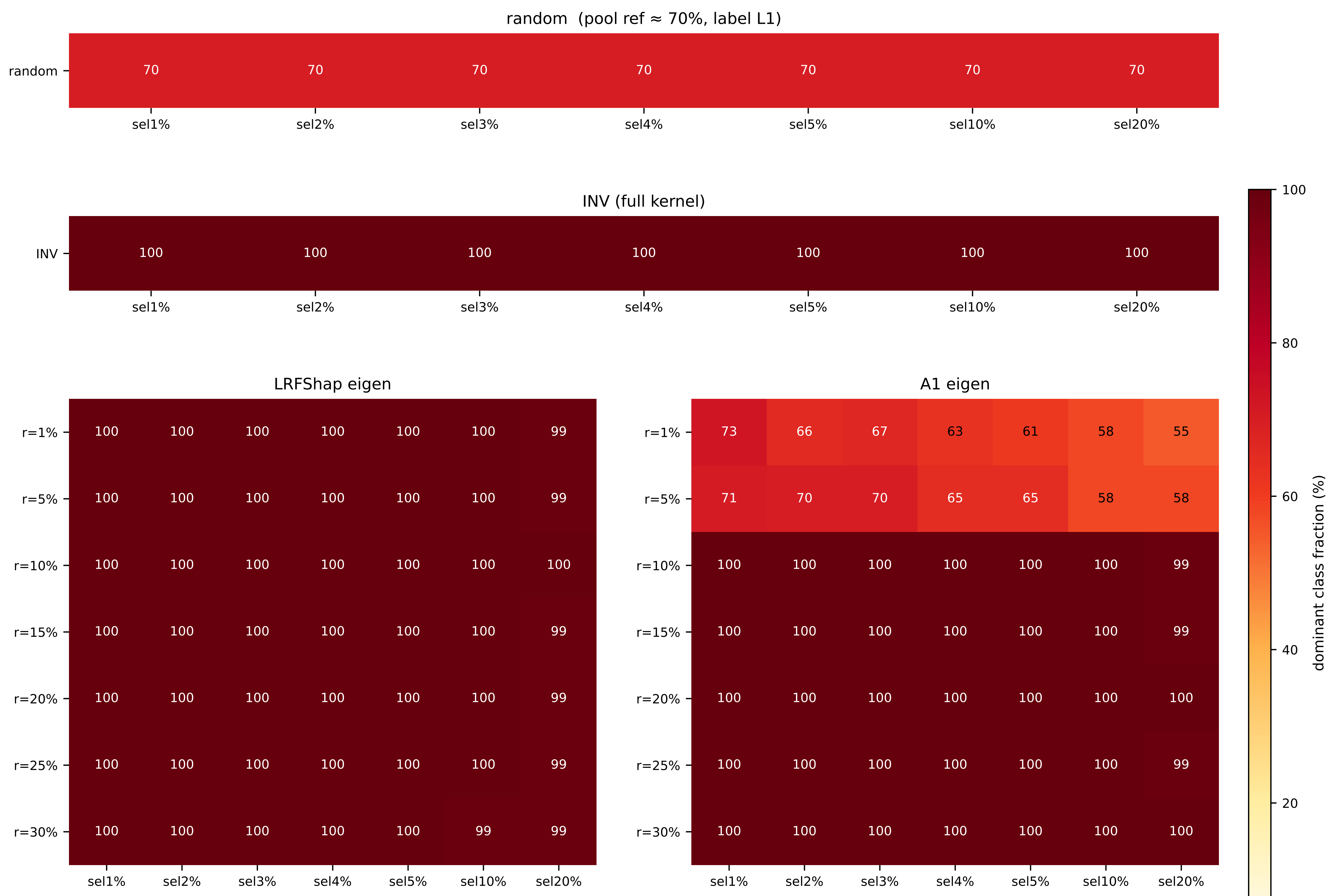


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

mr pos50 valimb500_pos70 — top-Shapley class composition

MR train 50/50 + val 70/30 label1-maj (n=4500, val=500)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70
LR / A1							
r=1%	L1=100 / L1=7	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=58	L1=99 / L1=55
r=5%	L1=100 / L1=7	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=58	L1=99 / L1=58
r=10%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=99
r=15%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=99 / L1=99
r=20%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=99 / L1=100
r=25%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=99 / L1=99
r=30%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=99 / L1=10	L1=99 / L1=100

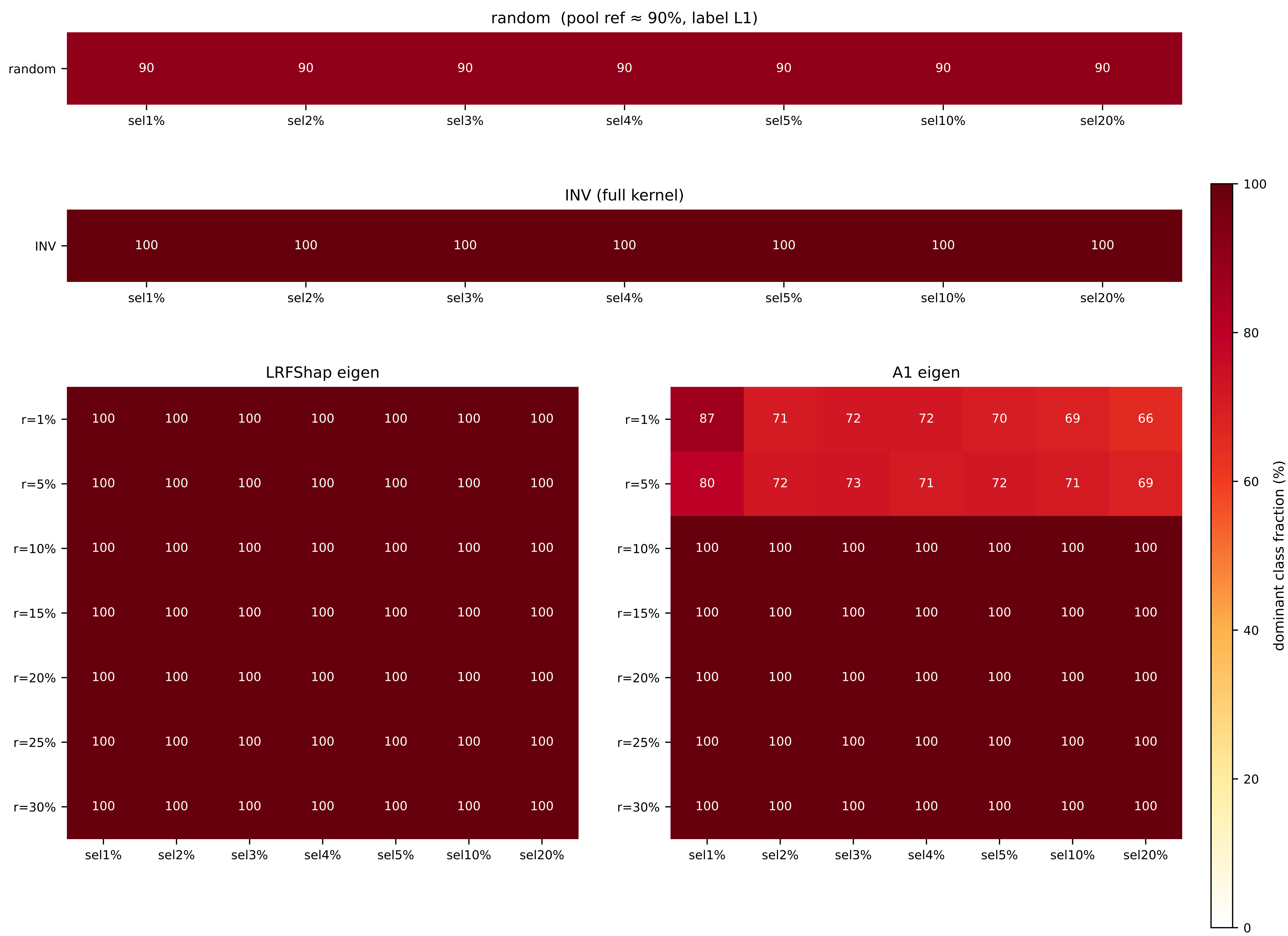


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

mr pos50 valimb500_pos90 — top-Shapley class composition

MR train 50/50 + val 90/10 label1-maj (n=4500, val=500)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=100 / L1=90	L1=100 / L1=90	L1=100 / L1=90	L1=100 / L1=90	L1=100 / L1=90	L1=100 / L1=90	L1=100 / L1=90
LR / A1							
r=1%	L1=100 / L1=8	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=70	L1=100 / L1=69	L1=100 / L1=66
r=5%	L1=100 / L1=80	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=69
r=10%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=100
r=15%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=100
r=20%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=100
r=25%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=100
r=30%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=100

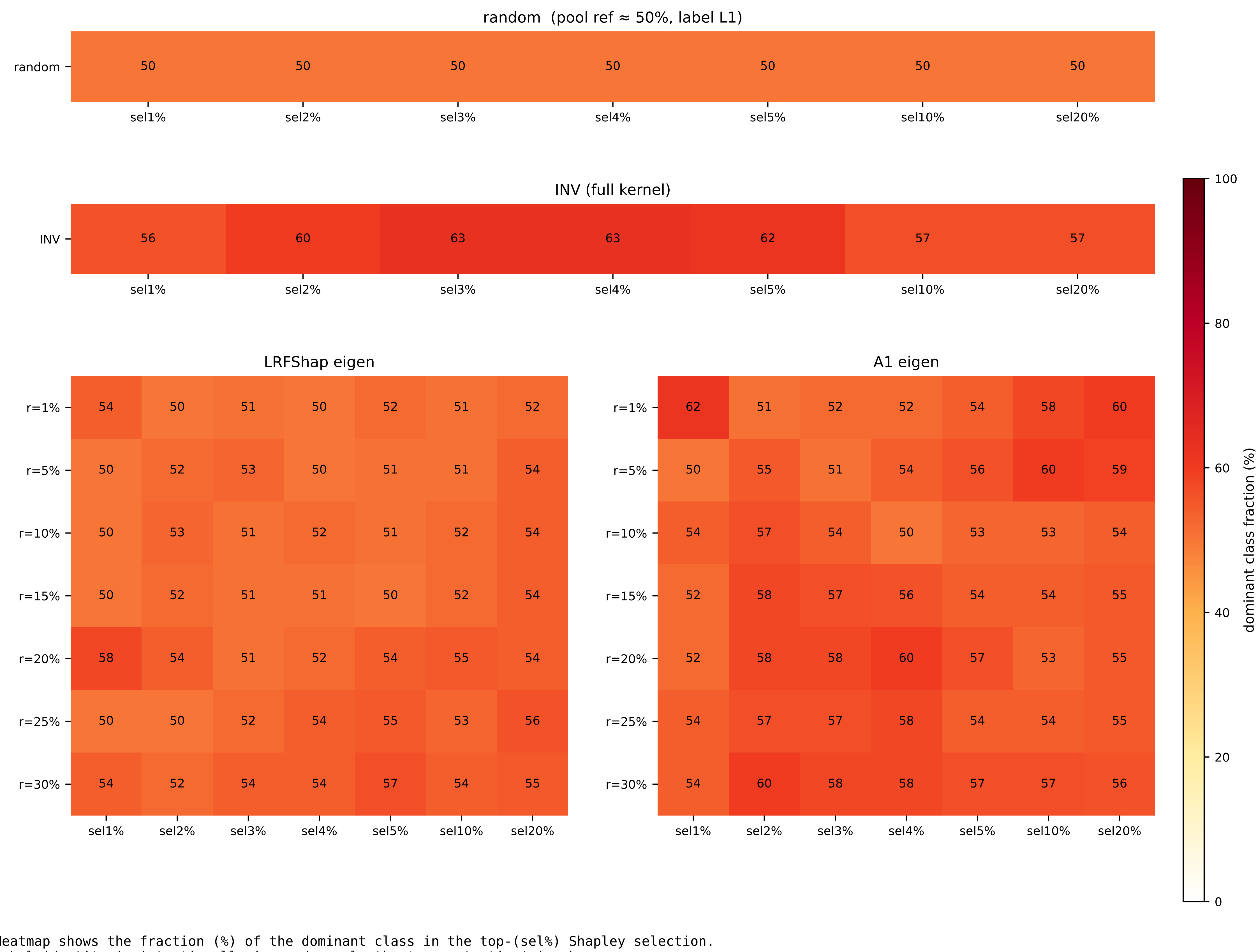


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

sst2 pos50 valbal856 — top-Shapley class composition

SST-2 train 50/50 + val balanced 50/50 (n=5000, val=856)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L0=56 / L1=50	L0=60 / L1=50	L0=63 / L1=50	L0=63 / L1=50	L0=62 / L1=50	L0=57 / L1=50	L0=57 / L1=50
LR / A1							
r=1%	L1=54 / L1=62	L1=50 / L0=51	L1=51 / L0=52	L1=50 / L0=52	L1=52 / L0=54	L0=51 / L0=58	L0=52 / L0=60
r=5%	L1=50 / L1=50	L1=52 / L0=55	L1=53 / L0=51	L0=50 / L0=54	L0=51 / L0=56	L0=51 / L0=60	L0=54 / L0=59
r=10%	L1=50 / L0=54	L1=53 / L0=57	L1=51 / L0=54	L0=52 / L0=50	L0=51 / L0=53	L0=52 / L0=53	L0=54 / L0=54
r=15%	L1=50 / L0=52	L1=52 / L0=58	L0=51 / L0=57	L0=51 / L0=56	L1=50 / L0=54	L0=52 / L0=54	L0=54 / L0=55
r=20%	L1=58 / L0=52	L1=54 / L0=58	L1=51 / L0=58	L0=52 / L0=60	L0=54 / L0=57	L0=55 / L0=53	L0=54 / L0=55
r=25%	L1=50 / L0=54	L1=50 / L0=57	L0=52 / L0=57	L0=54 / L0=58	L0=55 / L0=54	L0=53 / L0=54	L0=56 / L0=55
r=30%	L1=54 / L0=54	L1=52 / L0=60	L0=54 / L0=58	L0=54 / L0=58	L0=57 / L0=57	L0=54 / L0=57	L0=55 / L0=56

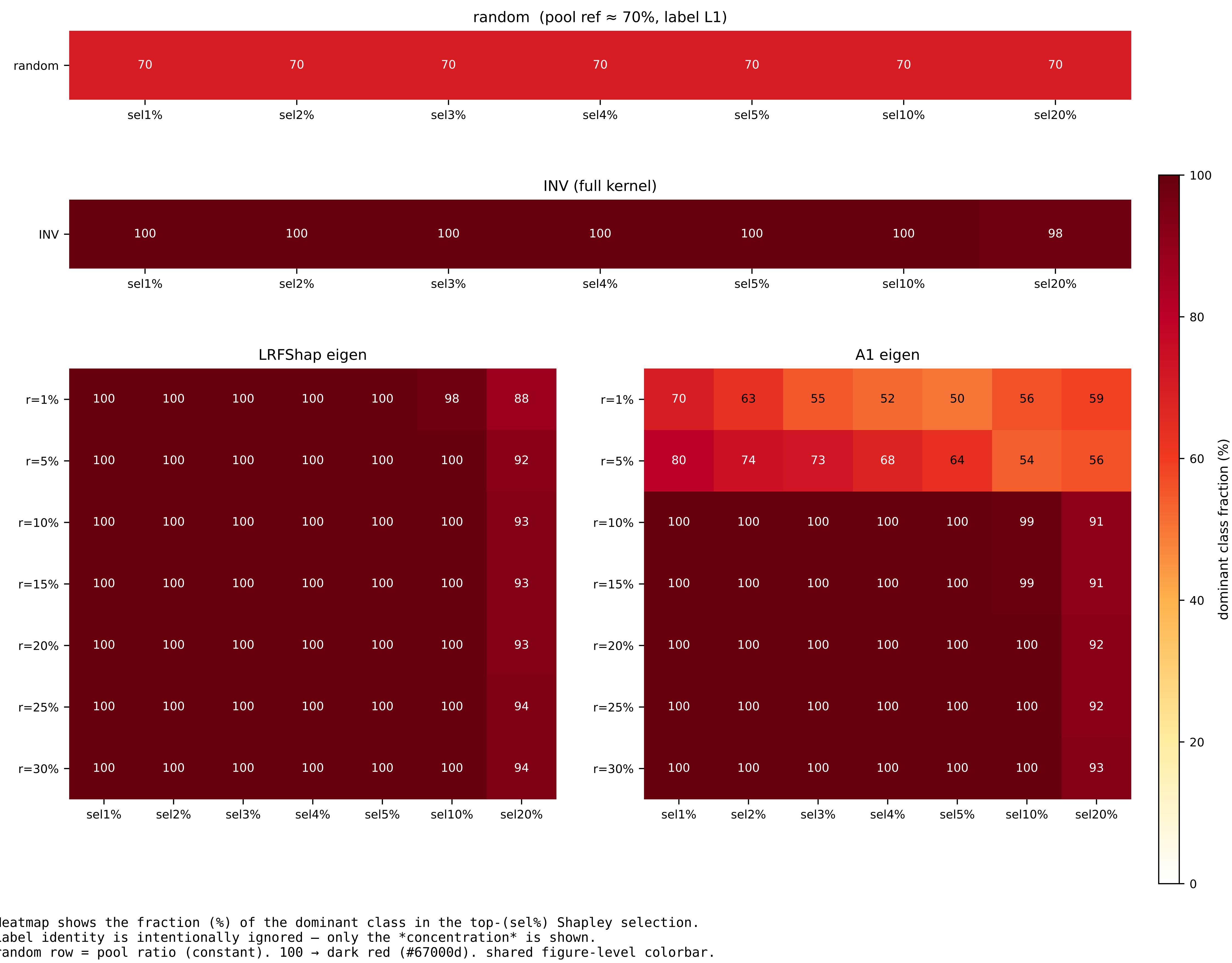



```
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

sst2 pos50 valimb400_pos70 — top-Shapley class composition

SST-2 train 50/50 + val 70/30 label1-maj (n=5000, val=400)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=98 / L1=70
LR / A1							
r=1%	L1=100 / L1=70	L1=100 / L1=6	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=50	L1=98 / L0=56	L1=88 / L0=59
r=5%	L1=100 / L1=80	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=68	L1=100 / L1=6	L1=100 / L1=54	L1=92 / L0=56
r=10%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=9	L1=93 / L1=91
r=15%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=9	L1=93 / L1=91
r=20%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=93 / L1=92
r=25%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=94 / L1=92
r=30%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=94 / L1=93

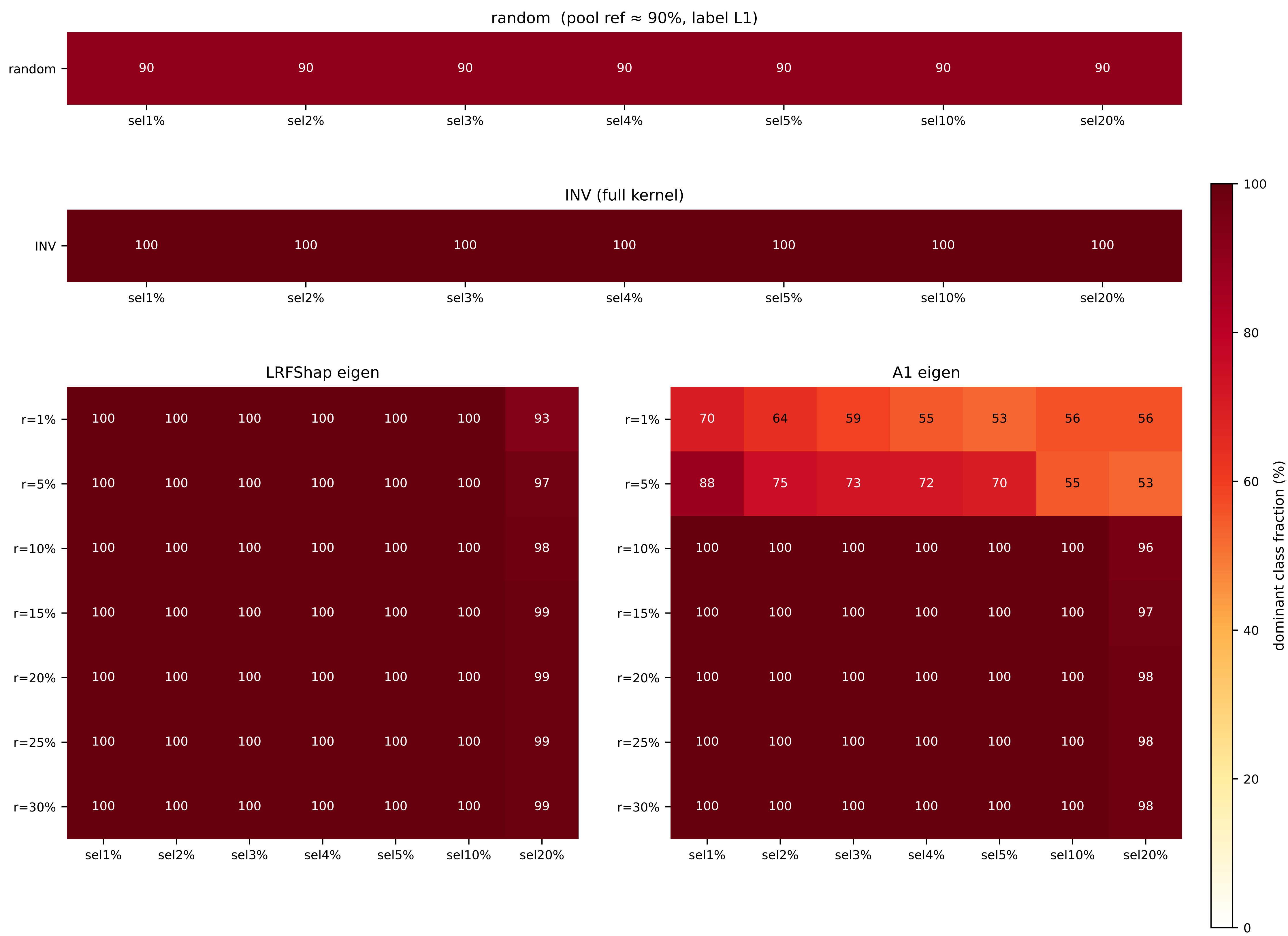


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

sst2 pos50 valimb400_pos90 — top-Shapley class composition

SST-2 train 50/50 + val 90/10 label1-maj (n=5000, val=400)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=100 / L1=9	1=100 / L1=9	1=100 / L1=9	1=100 / L1=9	1=100 / L1=9	1=100 / L1=9	L1=100 / L1=90
LR / A1							
r=1%	L1=100 / L1=7	1=100 / L1=6	1=100 / L1=5	1=100 / L1=5	1=100 / L1=5	1=100 / L0=5	L1=93 / L0=56
r=5%	L1=100 / L1=88	1=100 / L1=7	1=100 / L1=7	1=100 / L1=7	1=100 / L1=70	1=100 / L1=5	L1=97 / L0=53
r=10%	L1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	L1=98 / L1=96
r=15%	L1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	L1=99 / L1=97
r=20%	L1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	L1=99 / L1=98
r=25%	L1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	L1=99 / L1=98
r=30%	L1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	1=100 / L1=10	L1=99 / L1=98

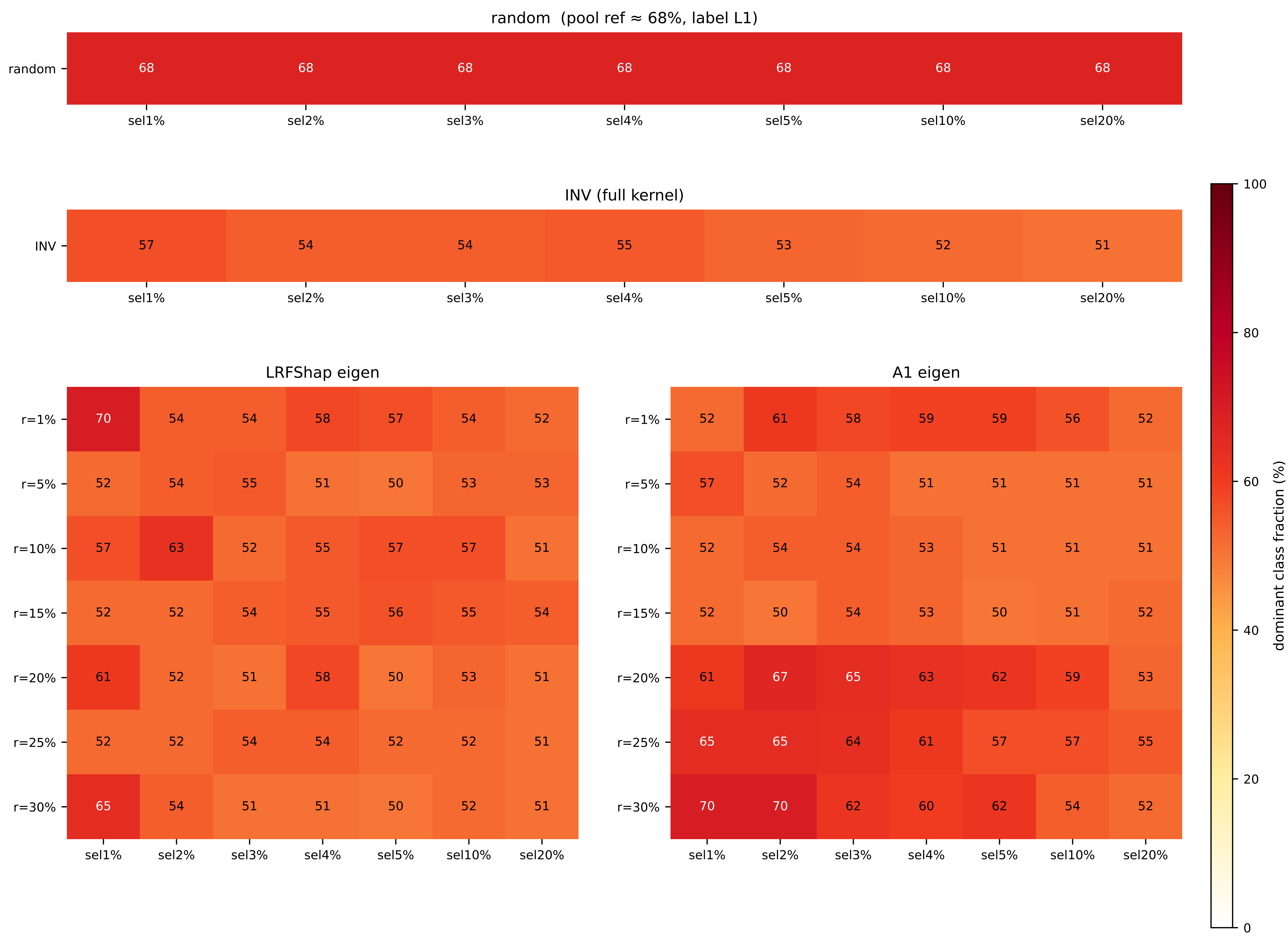



```
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

mrpc pos50 valbal258 — top-Shapley class composition

MRPC train 50/50 + val balanced 50/50 (n=2300, val=258)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L1=57 / L1=68	L1=54 / L1=68	L1=54 / L1=68	L1=55 / L1=68	L1=53 / L1=68	L1=52 / L1=68	L1=51 / L1=68
LR / A1							
r=1%	L0=70 / L1=52	L0=54 / L0=61	L0=54 / L0=58	L0=58 / L0=59	L0=57 / L0=59	L0=54 / L0=56	L0=52 / L0=52
r=5%	L0=52 / L0=57	L0=54 / L1=52	L0=55 / L1=54	L0=51 / L1=51	L0=50 / L1=51	L1=53 / L0=51	L1=53 / L1=51
r=10%	L1=57 / L0=52	L1=63 / L1=54	L1=52 / L1=54	L1=55 / L1=53	L1=57 / L1=51	L1=57 / L0=51	L1=51 / L1=51
r=15%	L0=52 / L0=52	L1=52 / L0=50	L1=54 / L1=54	L1=55 / L1=53	L1=56 / L1=50	L1=55 / L0=51	L1=54 / L1=52
r=20%	L1=61 / L1=61	L1=52 / L1=67	L1=51 / L1=65	L1=58 / L1=63	L1=50 / L1=62	L0=53 / L1=59	L0=51 / L1=53
r=25%	L1=52 / L1=65	L0=52 / L1=65	L0=54 / L1=64	L0=54 / L1=61	L0=52 / L1=57	L0=52 / L1=57	L1=51 / L1=55
r=30%	L1=65 / L1=70	L0=54 / L1=70	L0=51 / L1=62	L1=51 / L1=60	L0=50 / L1=62	L0=52 / L1=54	L1=51 / L1=52

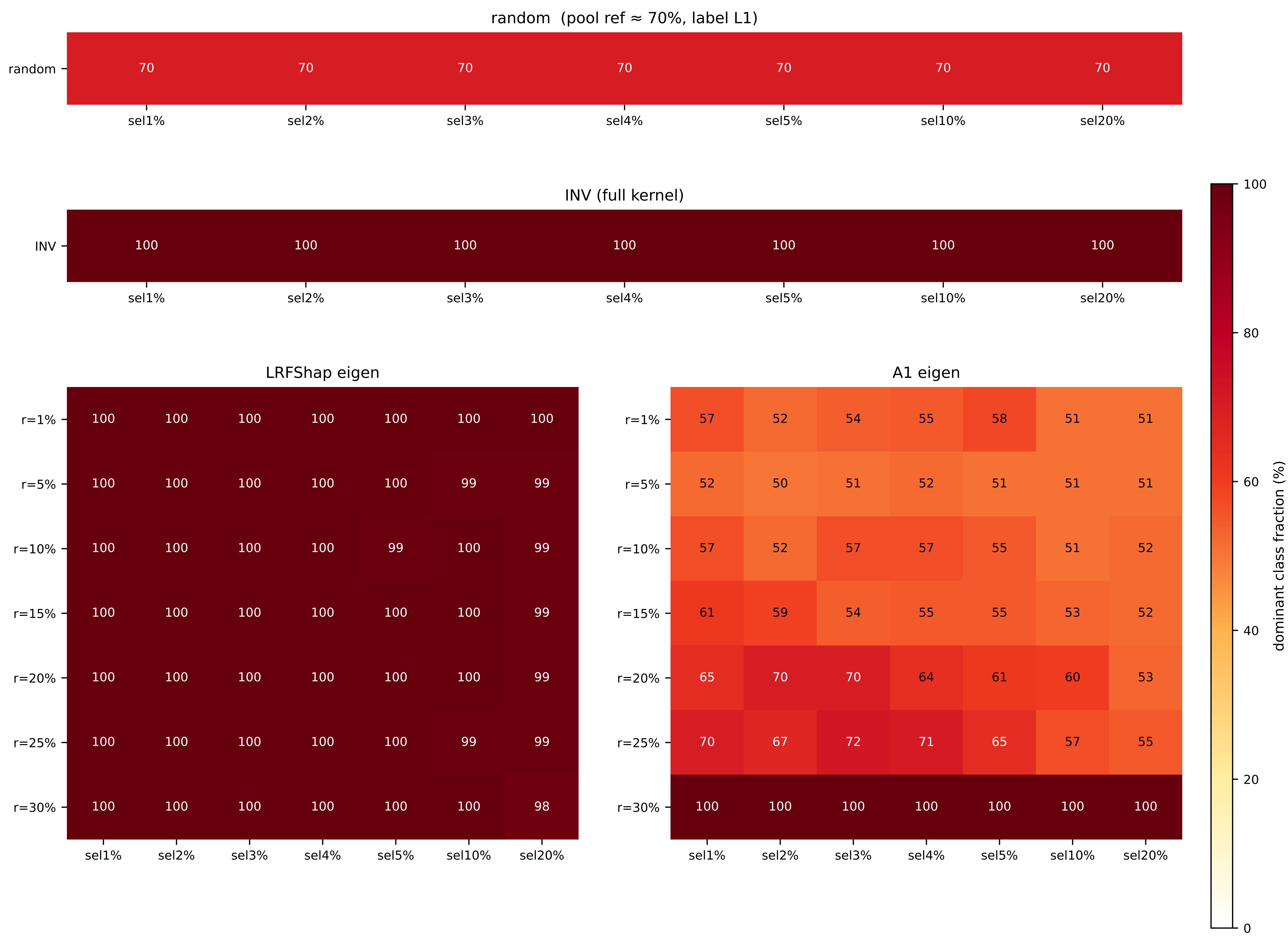


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

mrpc pos50 valimb300_pos70 — top-Shapley class composition

MRPC train 50/50 + val 70/30 label1-maj (n=2300, val=300)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=70
LR / A1							
r=1%	L1=100 / L0=5	L1=100 / L1=5	L1=100 / L0=5	L1=100 / L0=5	L1=100 / L0=5	L1=100 / L1=5	L1=100 / L1=51
r=5%	L1=100 / L0=5	L1=100 / L0=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L0=51	L1=99 / L0=51	L1=99 / L0=51
r=10%	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=57	L1=99 / L1=55	L1=100 / L1=51	L1=99 / L1=52
r=15%	L1=100 / L1=6	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=53	L1=99 / L1=52
r=20%	L1=100 / L1=6	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=60	L1=99 / L1=53
r=25%	L1=100 / L1=70	L1=100 / L1=6	L1=100 / L1=7	L1=100 / L1=7	L1=100 / L1=65	L1=99 / L1=57	L1=99 / L1=55
r=30%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=98 / L1=100

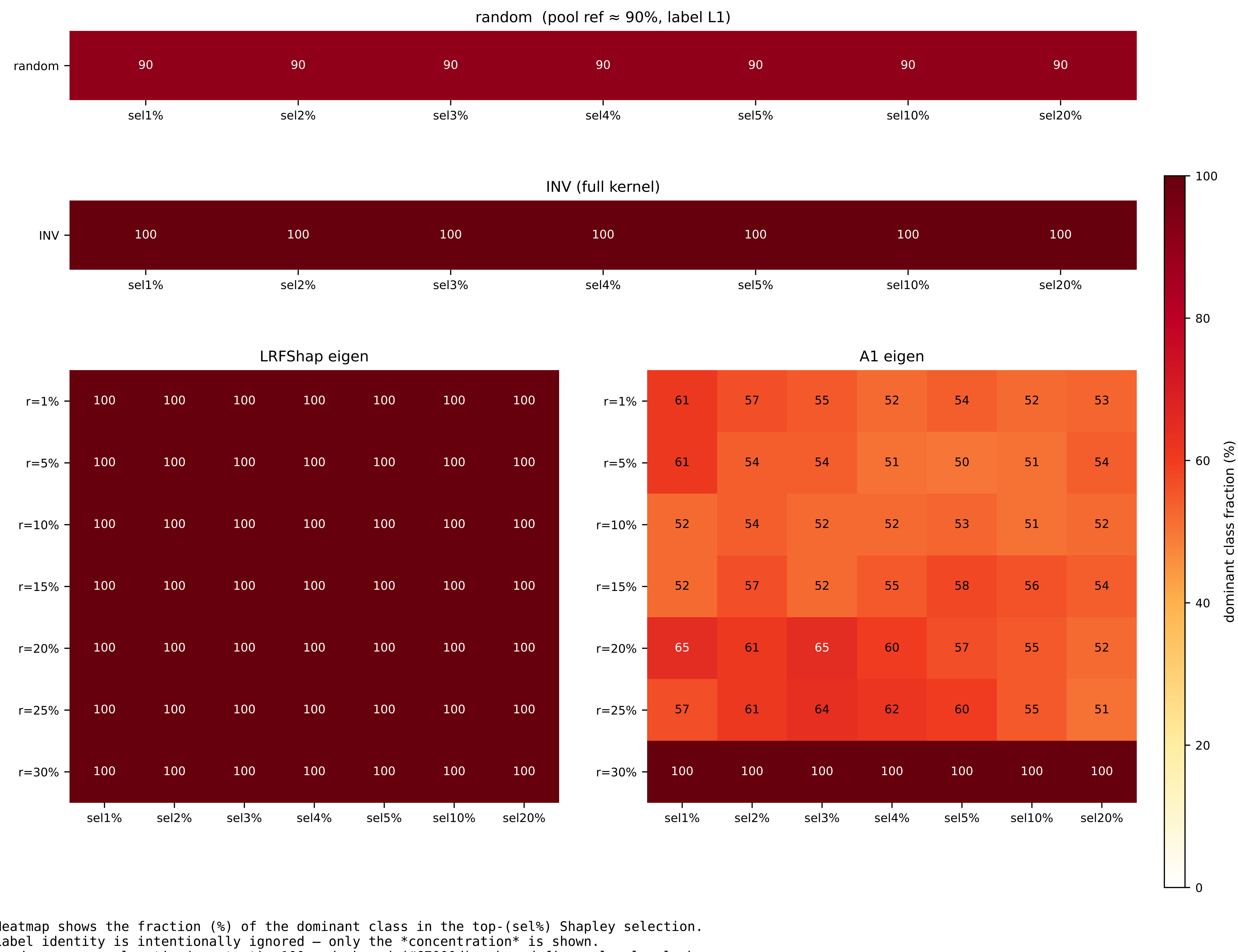


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

mrpc pos50 valimb300_pos90 — top-Shapley class composition

MRPC train 50/50 + val 90/10 label1-maj (n=2300, val=300)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=90
LR / A1							
r=1%	L1=100 / L0=6	L1=100 / L0=5	L1=100 / L0=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=53
r=5%	L1=100 / L0=6	L1=100 / L0=5	L1=100 / L0=5	L1=100 / L1=5	L1=100 / L1=50	L1=100 / L0=5	L1=100 / L1=54
r=10%	L1=100 / L0=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L0=5	L1=100 / L0=5	L1=100 / L0=5	L1=100 / L1=52
r=15%	L1=100 / L1=5	L1=100 / L0=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=54
r=20%	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=60	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=52
r=25%	L1=100 / L1=5	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=60	L1=100 / L1=5	L1=100 / L1=51
r=30%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=100

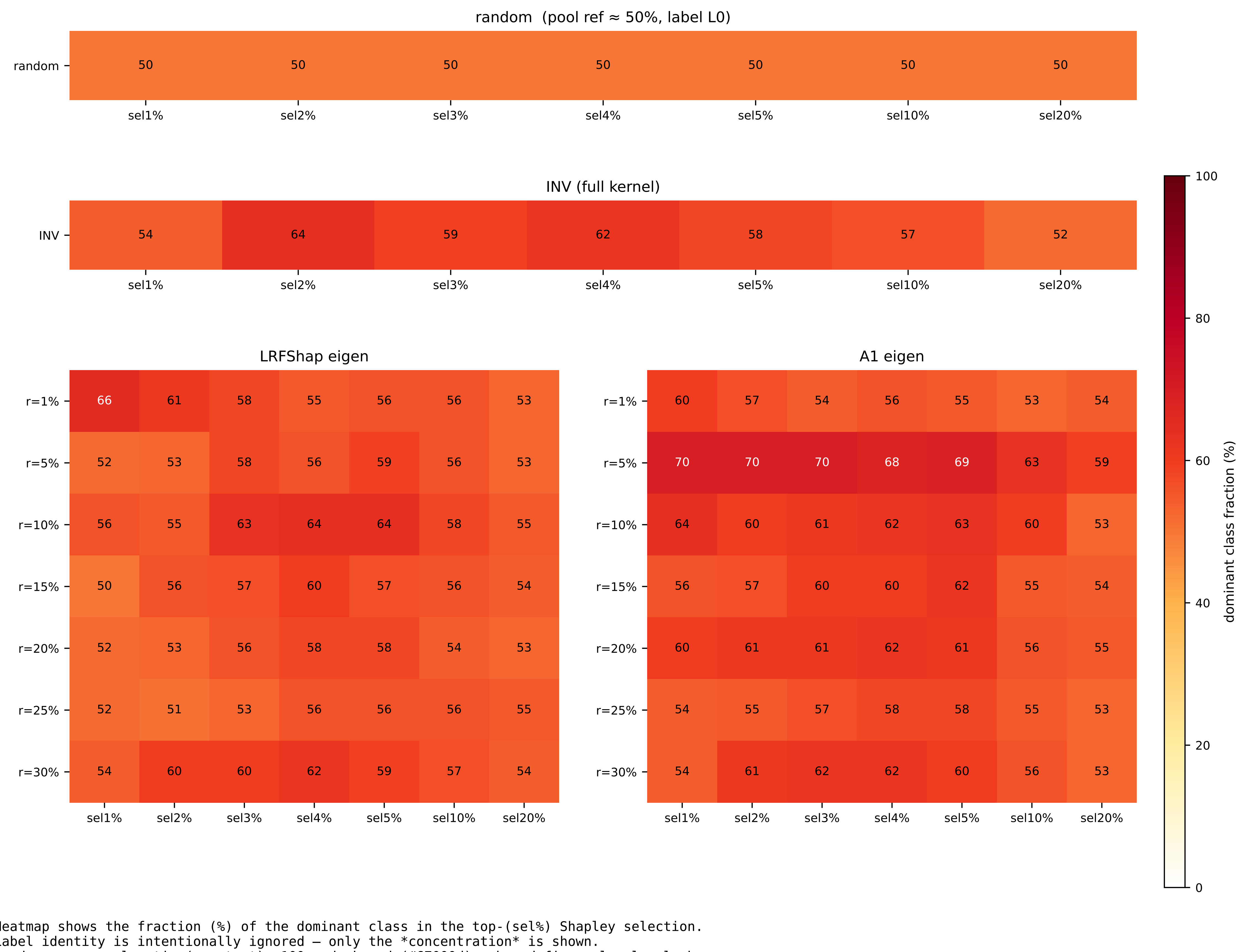



```
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

qqp pos50 valbal1000 — top-Shapley class composition

QQP train 50/50 + val balanced 50/50 (n=5000, val=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=54 / L0=50	L1=64 / L0=50	L1=59 / L0=50	L1=62 / L0=50	L1=58 / L0=50	L1=57 / L0=50	L1=52 / L0=50
LR / A1							
r=1%	L1=66 / L1=60	L1=61 / L1=57	L1=58 / L1=54	L1=55 / L1=56	L1=56 / L1=55	L1=56 / L1=53	L1=53 / L1=54
r=5%	L1=52 / L1=70	L1=53 / L1=70	L1=58 / L1=70	L1=56 / L1=68	L1=59 / L1=69	L1=56 / L1=63	L1=53 / L1=59
r=10%	L1=56 / L1=64	L1=55 / L1=60	L1=63 / L1=61	L1=64 / L1=62	L1=64 / L1=63	L1=58 / L1=60	L1=55 / L1=53
r=15%	L0=50 / L1=56	L1=56 / L1=57	L1=57 / L1=60	L1=60 / L1=60	L1=57 / L1=62	L1=56 / L1=55	L1=54 / L1=54
r=20%	L0=52 / L1=60	L1=53 / L1=61	L1=56 / L1=61	L1=58 / L1=62	L1=58 / L1=61	L1=54 / L1=56	L1=53 / L1=55
r=25%	L1=52 / L0=54	L1=51 / L1=55	L1=53 / L1=57	L1=56 / L1=58	L1=56 / L1=58	L1=56 / L1=55	L1=55 / L1=53
r=30%	L1=54 / L1=54	L1=60 / L1=61	L1=60 / L1=62	L1=62 / L1=62	L1=59 / L1=60	L1=57 / L1=56	L1=54 / L1=53

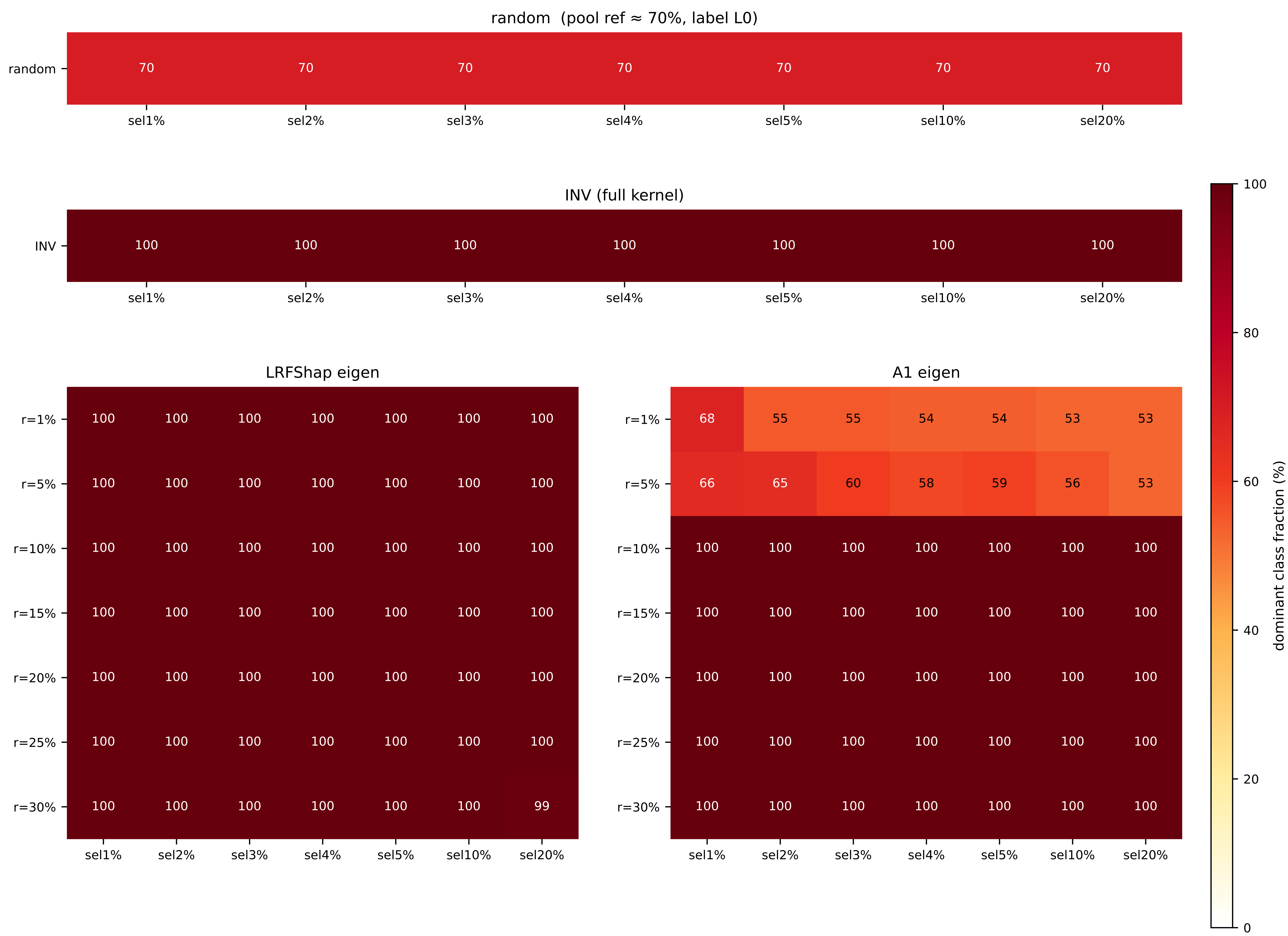


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

qqp pos50 valimb1000_pos30 — top-Shapley class composition

QQP train 50/50 + val 70/30 label0-maj (n=5000, val=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L0=70	L0=100 / L0=70	L0=100 / L0=70	L0=100 / L0=70	L0=100 / L0=70	L0=100 / L0=70	L0=100 / L0=70
LR / A1							
r=1%	L0=100 / L1=68	L0=100 / L1=55	L0=100 / L1=55	L0=100 / L1=54	L0=100 / L1=54	L0=100 / L1=53	L0=100 / L1=53
r=5%	L0=100 / L1=66	L0=100 / L1=65	L0=100 / L1=60	L0=100 / L1=58	L0=100 / L1=59	L0=100 / L1=56	L0=100 / L1=53
r=10%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=15%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=20%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=25%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=30%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=99 / L0=100

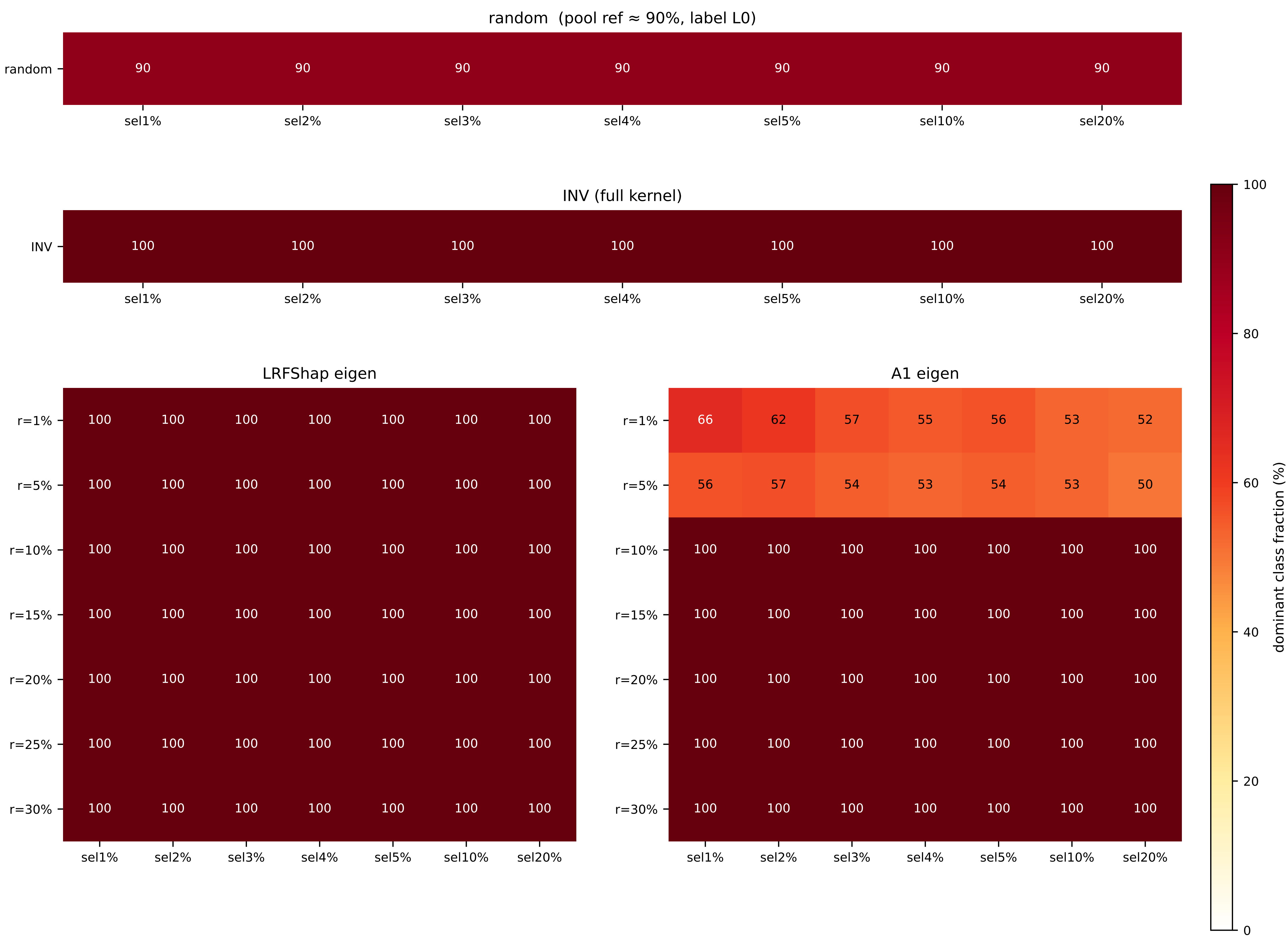



```
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

qqp pos50 valimb1000_pos10 — top-Shapley class composition

QQP train 50/50 + val 90/10 label0-maj (n=5000, val=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90
LR / A1							
r=1%	L0=100 / L1=66	L0=100 / L1=62	L0=100 / L1=57	L0=100 / L1=55	L0=100 / L1=56	L0=100 / L1=53	L0=100 / L1=52
r=5%	L0=100 / L1=56	L0=100 / L1=57	L0=100 / L1=54	L0=100 / L1=53	L0=100 / L1=54	L0=100 / L1=53	L0=100 / L1=50
r=10%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=15%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=20%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=25%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=30%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100

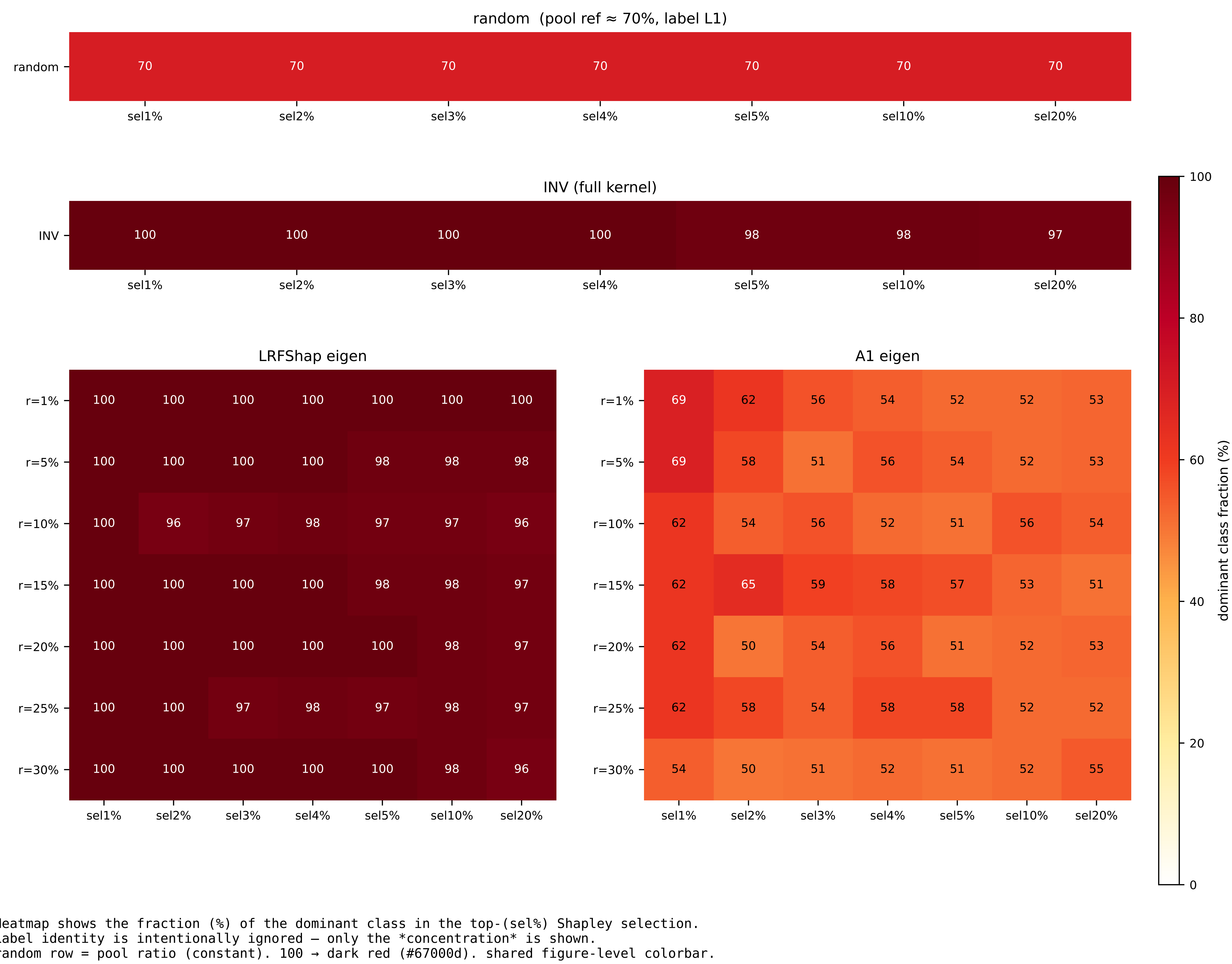


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

rte pos50 valimb145_pos70 — top-Shapley class composition

RTE train 50/50 + val 70/30 label1-maj (n=1300, val=145)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=100 / L1=70	L1=98 / L1=70	L1=98 / L1=70	L1=97 / L1=70
LR / A1							
r=1%	L1=100 / L0=69	L1=100 / L0=62	L1=100 / L0=56	L1=100 / L1=54	L1=100 / L1=52	L1=100 / L1=52	L1=100 / L1=53
r=5%	L1=100 / L0=69	L1=100 / L0=58	L1=100 / L0=51	L1=100 / L1=56	L1=98 / L1=54	L1=98 / L1=52	L1=98 / L1=53
r=10%	L1=100 / L1=62	L1=96 / L0=54	L1=97 / L1=56	L1=98 / L1=52	L1=97 / L1=51	L1=97 / L1=56	L1=96 / L1=54
r=15%	L1=100 / L1=62	L1=100 / L1=65	L1=100 / L1=59	L1=100 / L1=58	L1=98 / L1=57	L1=98 / L1=53	L1=97 / L1=51
r=20%	L1=100 / L1=62	L1=100 / L1=50	L1=100 / L1=54	L1=100 / L1=56	L1=100 / L1=51	L1=98 / L1=52	L1=97 / L1=53
r=25%	L1=100 / L1=62	L1=100 / L1=58	L1=97 / L1=54	L1=98 / L1=58	L1=97 / L1=58	L1=98 / L1=52	L1=97 / L0=52
r=30%	L1=100 / L1=54	L1=100 / L0=50	L1=100 / L1=51	L1=100 / L0=52	L1=100 / L1=51	L1=98 / L1=52	L1=96 / L1=55

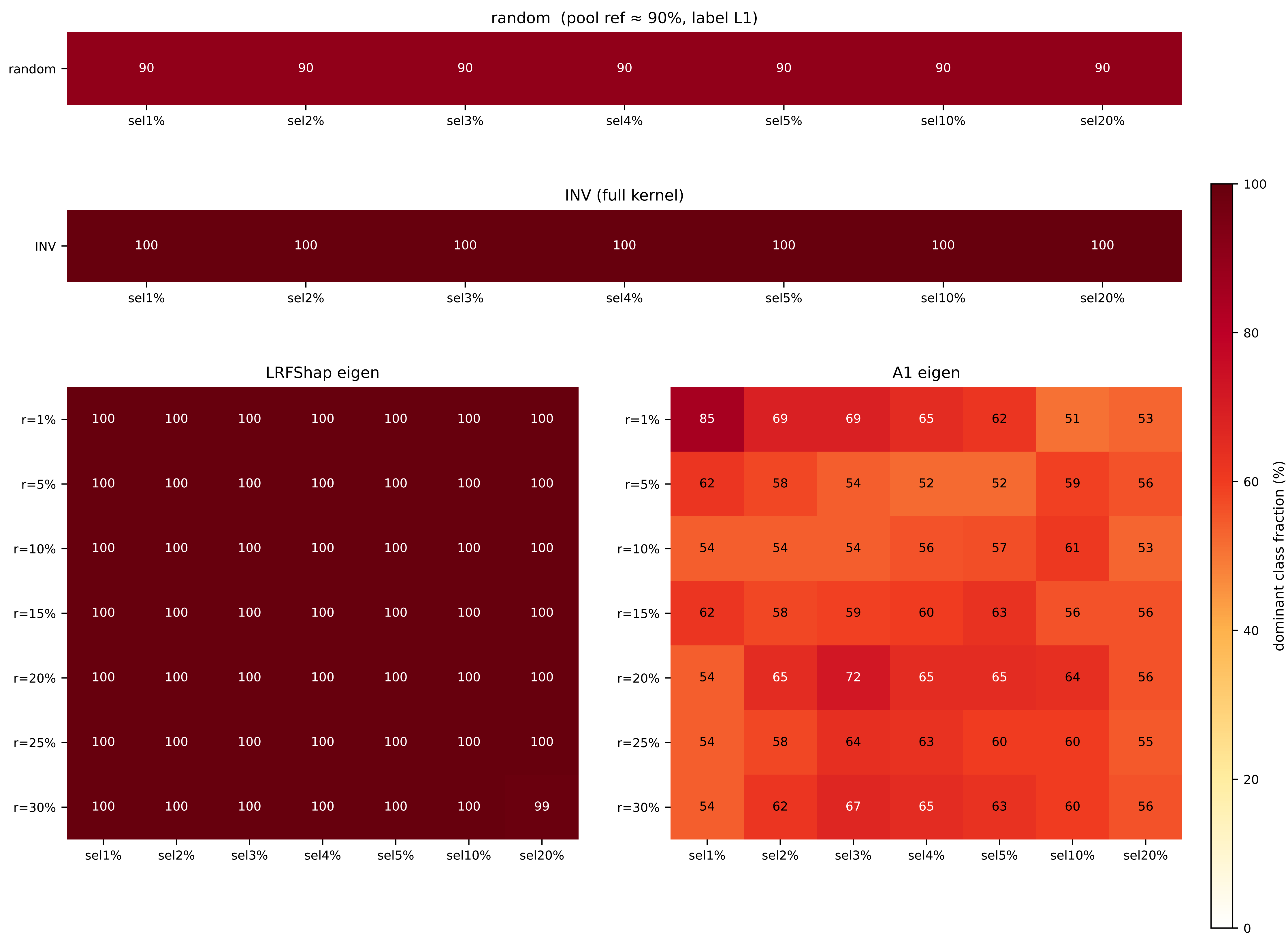



```
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

rte pos50 valimb145_pos90 — top-Shapley class composition

RTE train 50/50 + val 90/10 label1-maj (n=1300, val=145)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=9	L1=100 / L1=90
LR / A1							
r=1%	L1=100 / L0=8	L1=100 / L0=6	L1=100 / L0=6	L1=100 / L0=6	L1=100 / L0=6	L1=100 / L0=5	L1=100 / L1=53
r=5%	L1=100 / L0=6	L1=100 / L0=58	L1=100 / L0=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=56
r=10%	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L1=6	L1=100 / L1=53
r=15%	L1=100 / L1=6	L1=100 / L1=58	L1=100 / L1=5	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=5	L1=100 / L1=56
r=20%	L1=100 / L1=5	L1=100 / L1=6	L1=100 / L1=7	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=56
r=25%	L1=100 / L1=5	L1=100 / L1=58	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=55
r=30%	L1=100 / L1=5	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=100 / L1=6	L1=99 / L1=56

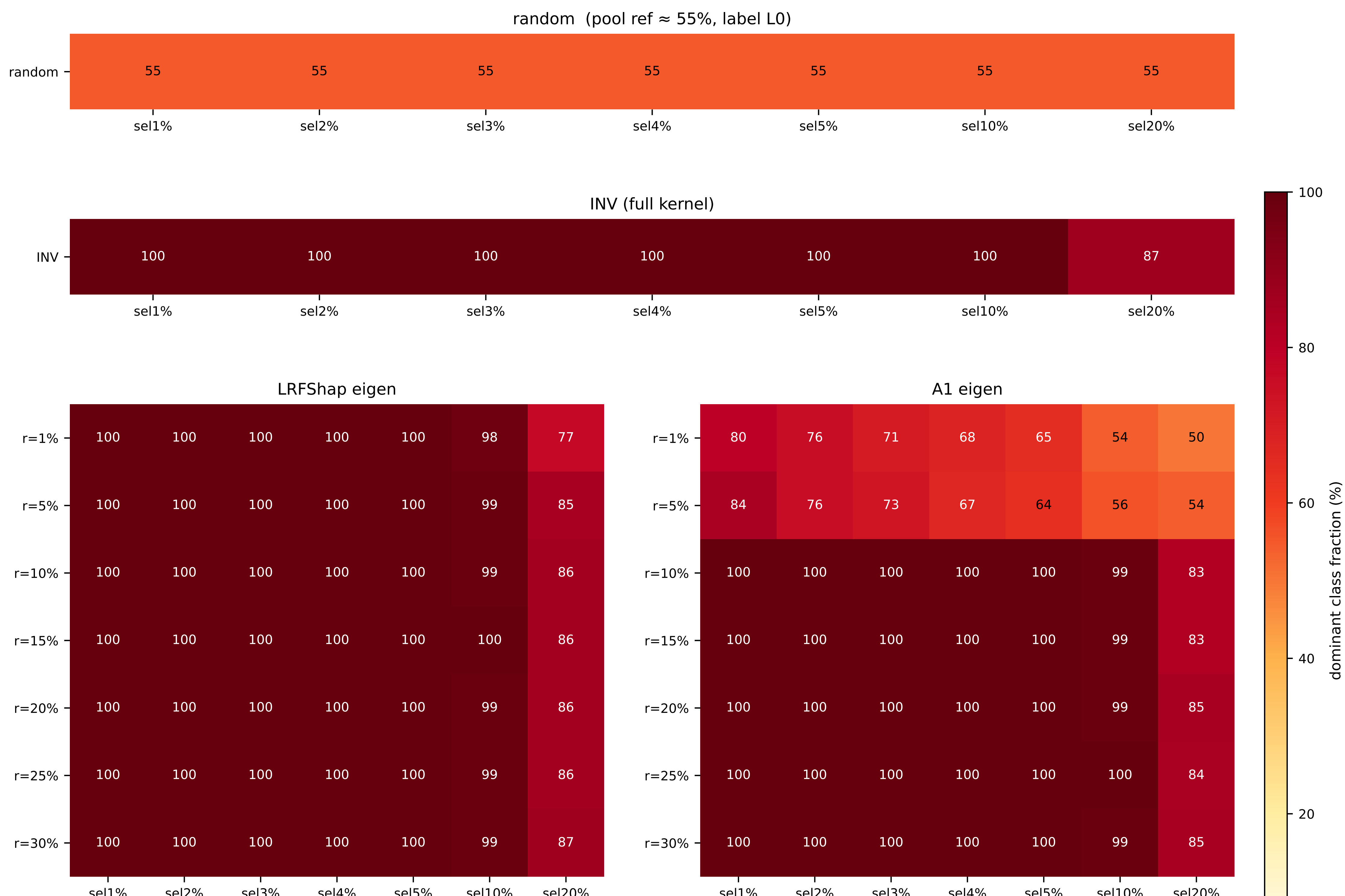


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

ag_news cls25_25_25_25 valimb1000_cls55_15_15_15 — top-Shapley class composition

AG News train cls25 + val cls55_15_15_15 label0-maj (n=5000, val=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L0=5	L0=100 / L0=5	L0=100 / L0=5	L0=100 / L0=5	L0=100 / L0=5	L0=100 / L0=55	L0=87 / L0=55
LR / A1							
r=1%	L0=100 / L0=8	L0=100 / L0=7	L0=100 / L0=7	L0=100 / L0=6	L0=100 / L0=6	L0=98 / L0=54	L0=77 / L0=50
r=5%	L0=100 / L0=8	L0=100 / L0=7	L0=100 / L0=7	L0=100 / L0=6	L0=100 / L0=64	L0=99 / L0=56	L0=85 / L0=54
r=10%	L0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	L0=99 / L0=99	L0=86 / L0=83
r=15%	L0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	L0=100 / L0=9	L0=86 / L0=83
r=20%	L0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	L0=99 / L0=99	L0=86 / L0=85
r=25%	L0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	L0=99 / L0=10	L0=86 / L0=84
r=30%	L0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	L0=99 / L0=99	L0=87 / L0=85

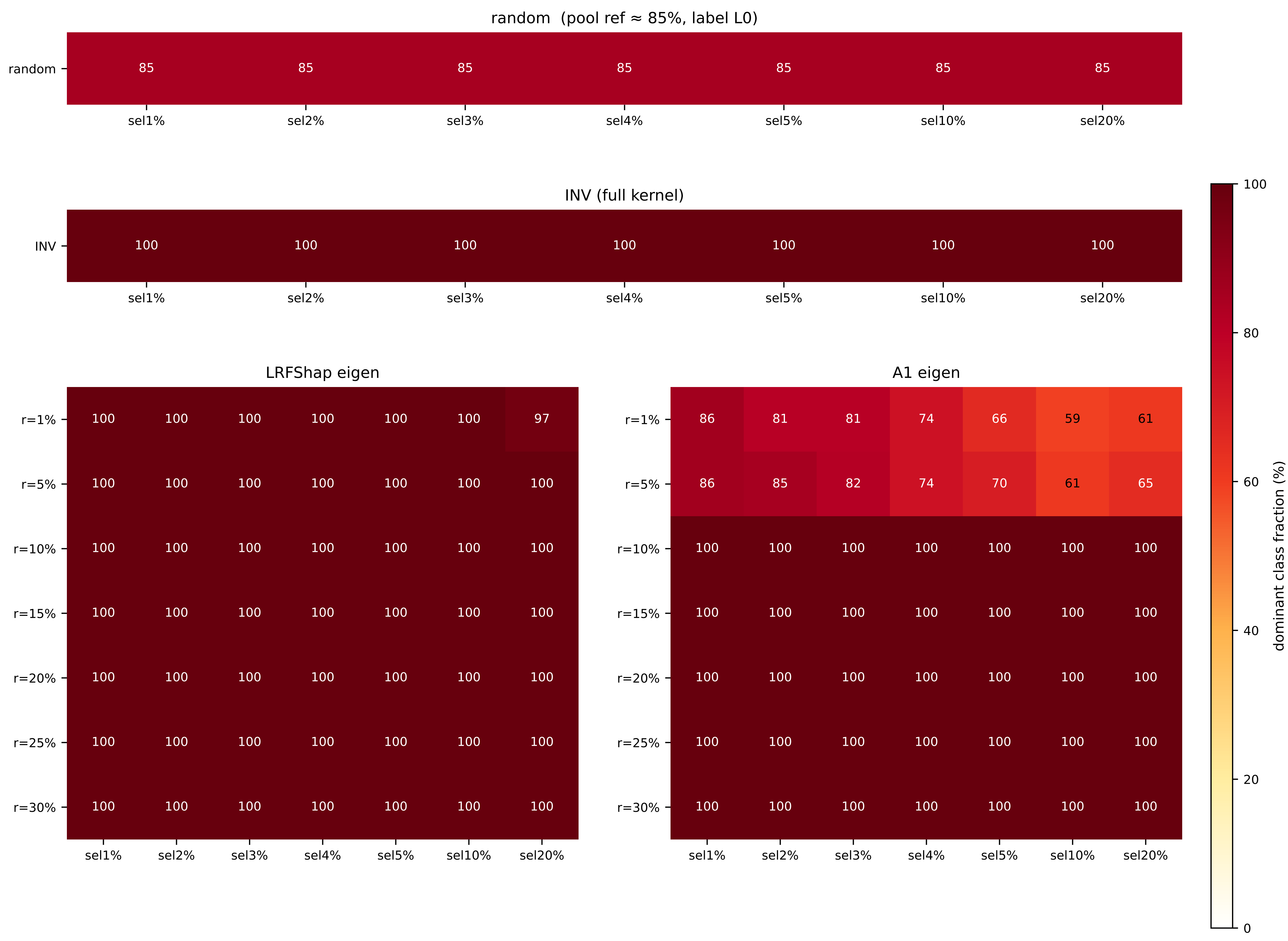



```
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

ag_news cls25_25_25_25 valimb1000_cls85_05_05_05 — top-Shapley class composition

AG News train cls25 + val cls85_05_05_05 label0-maj (n=5000, val=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L0=85	L0=100 / L0=85	L0=100 / L0=85	L0=100 / L0=85	L0=100 / L0=85	L0=100 / L0=85	L0=100 / L0=85
LR / A1							
r=1%	L0=100 / L0=86	L0=100 / L0=81	L0=100 / L0=81	L0=100 / L0=74	L0=100 / L0=66	L0=100 / L0=59	L0=97 / L0=61
r=5%	L0=100 / L0=86	L0=100 / L0=85	L0=100 / L0=82	L0=100 / L0=74	L0=100 / L0=70	L0=100 / L0=61	L0=100 / L0=65
r=10%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=15%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=20%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=25%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100
r=30%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100

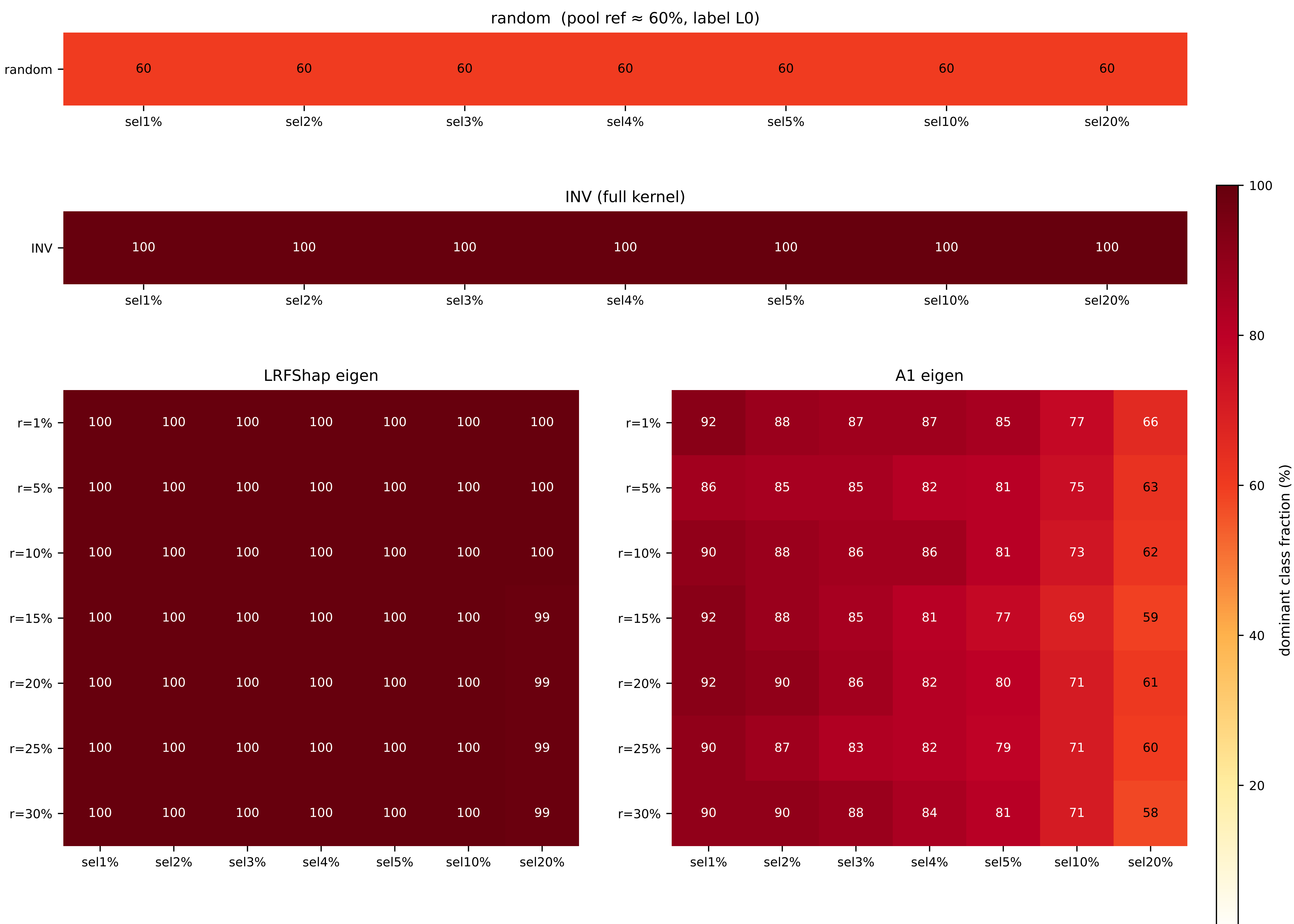



```
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

mnli cls33_33_33 valimb1000_cls60_20_20 — top-Shapley class composition

MNLI train cls33 + val cls60_20_20 label0-maj (n=5000, val=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L0=60	L0=100 / L0=60	L0=100 / L0=60	L0=100 / L0=60	L0=100 / L0=60	L0=100 / L0=60	L0=100 / L0=60
LR / A1							
r=1%	L0=100 / L0=92	L0=100 / L0=88	L0=100 / L0=87	L0=100 / L0=87	L0=100 / L0=85	L0=100 / L0=77	L0=100 / L0=66
r=5%	L0=100 / L0=86	L0=100 / L0=85	L0=100 / L0=85	L0=100 / L0=82	L0=100 / L0=81	L0=100 / L0=75	L0=100 / L0=63
r=10%	L0=100 / L0=90	L0=100 / L0=88	L0=100 / L0=86	L0=100 / L0=86	L0=100 / L0=81	L0=100 / L0=73	L0=100 / L0=62
r=15%	L0=100 / L0=92	L0=100 / L0=88	L0=100 / L0=85	L0=100 / L0=81	L0=100 / L0=77	L0=100 / L0=69	L0=99 / L0=59
r=20%	L0=100 / L0=92	L0=100 / L0=90	L0=100 / L0=86	L0=100 / L0=82	L0=100 / L0=80	L0=100 / L0=71	L0=99 / L0=61
r=25%	L0=100 / L0=90	L0=100 / L0=87	L0=100 / L0=83	L0=100 / L0=82	L0=100 / L0=79	L0=100 / L0=71	L0=99 / L0=60
r=30%	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=88	L0=100 / L0=84	L0=100 / L0=81	L0=100 / L0=71	L0=99 / L0=58

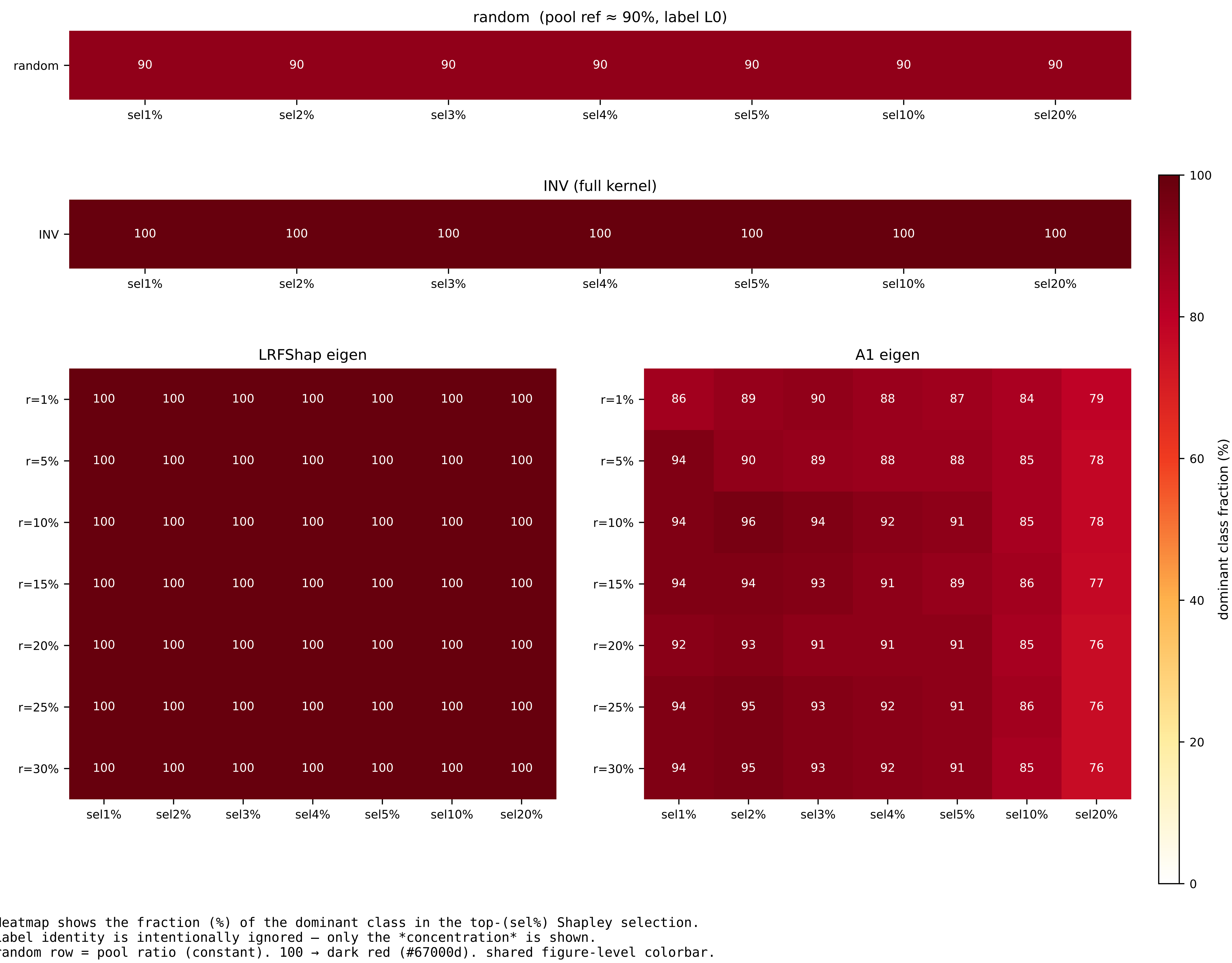


Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

mnli cls33_33_33 valimb1000_cls90_05_05 — top-Shapley class composition

MNLI train cls33 + val cls90_05_05 label0-maj (n=5000, val=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90	L0=100 / L0=90
LR / A1							
r=1%	L0=100 / L0=86	L0=100 / L0=89	L0=100 / L0=90	L0=100 / L0=88	L0=100 / L0=87	L0=100 / L0=84	L0=100 / L0=79
r=5%	L0=100 / L0=94	L0=100 / L0=90	L0=100 / L0=89	L0=100 / L0=88	L0=100 / L0=88	L0=100 / L0=85	L0=100 / L0=78
r=10%	L0=100 / L0=94	L0=100 / L0=96	L0=100 / L0=94	L0=100 / L0=92	L0=100 / L0=91	L0=100 / L0=85	L0=100 / L0=78
r=15%	L0=100 / L0=94	L0=100 / L0=94	L0=100 / L0=93	L0=100 / L0=91	L0=100 / L0=89	L0=100 / L0=86	L0=100 / L0=77
r=20%	L0=100 / L0=92	L0=100 / L0=93	L0=100 / L0=91	L0=100 / L0=91	L0=100 / L0=91	L0=100 / L0=85	L0=100 / L0=76
r=25%	L0=100 / L0=94	L0=100 / L0=95	L0=100 / L0=93	L0=100 / L0=92	L0=100 / L0=91	L0=100 / L0=86	L0=100 / L0=76
r=30%	L0=100 / L0=94	L0=100 / L0=95	L0=100 / L0=93	L0=100 / L0=92	L0=100 / L0=91	L0=100 / L0=85	L0=100 / L0=76




```
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

Shapley wall-clock time and speedup (= INV time / method time)

setting (node, GPU)	INV	LR r=1%	LR r=5%	LR r=10%	LR r=15%	LR r=20%	LR r=25%	LR r=30%	A1 r=1%	A1 r=5%	A1 r=10%	A1 r=15%	A1 r=20%	A1 r=25%	A1 r=30%
mr/pos50/valbal1000 (node05, RTX A6000)	6.8 h 1.0x	9.4 min 43.2x	23.7 min 17.2x	35.5 min 11.5x	55.6 min 7.3x	1.0 h 6.8x	1.3 h 5.2x	1.6 h 4.4x	9.4 min 43.3x	15.1 min 27.1x	16.8 min 24.2x	20.5 min 19.8x	23.4 min 17.4x	26.4 min 15.5x	28.4 min 14.4x
mr/pos50/valimb500_pos70 (node04, RTX 3090)	5.0 h 1.0x	10.5 min 28.6x	24.4 min 12.3x	37.9 min 7.9x	1.0 h 5.0x	1.2 h 4.1x	1.4 h 3.6x	1.8 h 2.8x	5.9 min 51.1x	12.5 min 24.1x	15.0 min 20.0x	16.3 min 18.5x	19.5 min 15.4x	22.0 min 13.7x	24.1 min 12.5x
mr/pos50/valimb500_pos90 (node04, RTX 3090)	9.3 h 1.0x	9.8 min 57.2x	20.1 min 27.8x	23.4 min 24.0x	39.8 min 14.1x	43.7 min 12.8x	55.1 min 10.2x	1.1 h 8.3x	6.8 min 82.2x	10.8 min 51.9x	12.3 min 45.6x	16.5 min 34.0x	15.9 min 35.1x	16.7 min 33.5x	18.7 min 30.0x
sst2/pos50/valbal856 (node04, RTX 3090)	3.5 h 1.0x	6.6 min 31.5x	21.8 min 9.5x	30.6 min 6.8x	38.8 min 5.4x	1.0 h 3.3x	1.1 h 3.3x	1.4 h 2.5x	4.3 min 48.3x	11.8 min 17.6x	13.7 min 15.2x	18.8 min 11.1x	19.8 min 10.5x	21.7 min 9.6x	28.4 min 7.3x
sst2/pos50/valimb400_pos70 (node04, RTX 3090) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
sst2/pos50/valimb400_pos90 (node04, RTX 3090) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
mrpc/pos50/valbal258 (node01, RTX 3090)	3.2 h 1.0x	5.2 min 36.1x	13.9 min 13.6x	19.8 min 9.6x	21.2 min 9.0x	23.4 min 8.1x	24.8 min 7.7x	27.1 min 7.0x	5.8 min 32.5x	8.1 min 23.5x	8.8 min 21.4x	10.7 min 17.8x	9.9 min 19.2x	10.1 min 18.7x	13.7 min 13.9x
mrpc/pos50/valimb300_pos70 (node04, RTX 3090) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
mrpc/pos50/valimb300_pos90 (node05, RTX A6000) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
qqp/pos50/valbal1000 (node05, RTX A6000)	19.2 h 1.0x	22.2 min 51.7x	32.2 min 35.6x	51.1 min 22.5x	1.2 h 16.0x	1.8 h 10.8x	1.9 h 10.3x	1.9 h 9.9x	15.3 min 75.3x	20.4 min 56.2x	29.3 min 39.3x	33.5 min 34.3x	41.2 min 27.9x	50.0 min 23.0x	57.7 min 19.9x
qqp/pos50/valimb1000_pos30 (node01, RTX 2080 Ti) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
qqp/pos50/valimb1000_pos10 (node03, RTX 3090) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
rte/pos50/valimb145_pos70 (node03, RTX 3090) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
rte/pos50/valimb145_pos90 (node03, RTX 3090) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ag_news/cls25_25_25_25/valimb1000_cls55_15_15_15 (node05, RTX A6000) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ag_news/cls25_25_25_25/valimb1000_cls85_05_05_05 (node05, RTX A6000) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
mnli/cls33_33_33/valimb1000_cls60_20_20 (node04, RTX A6000) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
mnli/cls33_33_33/valimb1000_cls90_05_05 (node03+node01+node05, RTX 3090 / RTX 2080Ti / (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

```
Cell format: 'wall-clock time / speedup' (speedup = INV time / method time at same setting).
INV column is the FreeShap baseline (full kernel inverse); its speedup is 1.0x by definition.
Time is the elapsed portion of the final tqdm '500/500 [HH:MM:SS<00:00, ...]' line
captured for each banner-tagged sub-run in the experiment logs.
Hardware labels reflect the node where each setting was actually run; speedups are NOT
comparable across rows because different GPUs were used.
Timing is independent of the accuracy metric — these numbers do not change between
the naive-acc and balanced-acc editions of this PDF.
```